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Boosting self-supervised multi-frame depth estimation with hybrid geometric-semantic constraints

Jiaojiao Fang 

School of Electronics and Information Engineering, Xi'an Jiaotong University, No. 28, Xianning West Road, Xi'an City, Shaanxi Province, 710049, China

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ABSTRACT

Self-supervised multi-frame monocular depth estimation leverages semantic appearance and geometric matching information to improve depth prediction through temporal cues, yet it continues to face challenges posed by semantic and geometric inconsistencies in low-texture, occluded, and dynamic regions. While existing methods often rely on spatial context to address such inconsistencies, they typically overlook explicit geometric priors. To overcome this limitation, we introduce an adaptive hybrid geometric and context-aware attention module, which refines corrupted cost volumes by aggregating local geometric and semantic information and propagating consistent neighboring costs guided by surface normals and epipolar geometry. Complementing this design, we propose an iterative multi-scale hybrid decoder that effectively fuses both multi-frame and single-frame cues encoded in the cost volumes. Extensive experiments across diverse real-world scenarios, including outdoor driving datasets (KITTI, Cityscapes), indoor environments (NYUv2, ScanNet), and challenging adverse-weather conditions (DENSE), validate the robustness, accuracy, and strong generalization capability of the proposed method.

1. Introduction

Monocular depth estimation aims to reconstruct the 3D structure of a scene from a single RGB image, a task crucial for applications such as 3D reconstruction (Luo et al., 2020), autonomous driving (Geiger et al., 2012), and augmented reality (Newcombe et al., 2011). While specialized depth sensors like LiDAR or structured-light systems can provide accurate measurements, using a single RGB camera offers a more cost-effective and scalable alternative. In recent years, self-supervised methods that learn from unlabeled video sequences have gained popularity, as they eliminate the need for expensive ground-truth depth annotations during training (Garg et al., 2016; Godard et al., 2017; Gordon et al., 2019; Zhou et al., 2017). These methods typically infer depth from monocular cues, such as texture variations and semantic object features (Yin et al., 2022); however, their accuracy still lags behind systems using dedicated depth sensors or multi-view learning (Li et al., 2023; Smolyanskiy et al., 2018).

To address this gap, recent multi-frame methods have incorporated explicit geometric reasoning by constructing cost volumes through feature-map warping across adjacent frames, inspired by Multi-View Stereo (MVS) principles (Hou et al., 2019; Im et al., 2019; Watson et al., 2021). These approaches aggregate multi-frame geometric

relations as matching costs, typically structured in 3D cost volumes (Feng et al., 2022; Long et al., 2022; Watson et al., 2021) or 2D cost maps (Gu et al., 2023), leading to improved depth accuracy. The 3D cost volume, in particular, has proven to be an efficient representation, capturing multi-frame matching probabilities over discrete depth hypotheses.

Despite the advantages of cost-volume-based multi-frame methods, there are still key limitations. Firstly, in dynamic, low-texture, or occluded regions, matching costs often exhibit local geometric and semantic inconsistencies that degrade the quality of the cost volume. Photometric matching deteriorates in these regions, leading to corrupted volumes with unreliable depth estimates. Secondly, current network architectures for fusing single-frame and multi-frame cues struggle to consistently integrate complementary information, especially in challenging regions with dynamic objects or weak textures. Existing fusion strategies fail to adaptively resolve conflicts between geometrically noisy multi-view cues and semantically rich but geometrically ambiguous single-frame cues. While several self-supervised methods have attempted to mitigate such local inconsistencies, many overlook the integration of geometric priors, which could directly resolve structural ambiguities in the cost volume. Previous approaches tackle these challenges individually: CAT-Net (Wu et al., 2024) introduces spatial

E-mail address: 995541569@qq.com.

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attention for cost regularization, IterDepth (Feng et al., 2024) employs recurrent geometry refinement, and ProDepth (Woo et al., 2024) focuses on late fusion of monocular and multi-view predictions. However, these methods tend to treat cost-volume rectification and cross-cue fusion as separate or sequential stages, without a unified mechanism that allows these processes to mutually guide and correct each other based on both geometric plausibility and semantic consistency.

To address these challenges, we propose a novel self-supervised multi-frame depth estimation framework that introduces two synergistic innovations: geometry- and semantics-aware cost aggregation, and iterative, attention-guided cross-cue fusion. Unlike traditional methods that rely on passively filtering noisy cost volumes, our hybrid aggregation module actively incorporates high-level semantic context and explicit geometric priors to detect and correct inconsistent matching costs. This not only improves geometric coherence but also enhances semantic consistency before fusion. Moreover, we introduce an iterative multi-scale architecture where single-frame (semantic) and multi-frame (geometric) cues interact through cross-attention mechanisms. This allows the progressive disambiguation of cost volumes with semantic guidance, while refining semantic features using geometric constraints. Unlike existing methods, the proposed approach creates a mutual refinement loop that integrates cost-volume purification with cross-cue fusion, enabling the model to reason about scene semantics and multi-view alignment jointly at each stage.

The primary contributions of the proposed method are:

- i) Geometry- and Semantics-Aware Adaptive Cost Aggregation: Unlike previous methods that address geometric consistency post hoc or rely on spatial filtering, we introduce a hybrid aggregation module that injects geometric priors (e.g., surface normals, epipolar constraints) and high-level semantic context into the cost filtering process. This approach enables consistent propagation of geometric-semantic information, distinguishing it from CAT-Net's fixed spatial attention or ProDepth's late-stage geometric regularization. By adaptively rectifying inconsistencies at the cost-volume level, the proposed method produces a more coherent and semantically grounded evidence map, especially in dynamic, occluded, and low-texture regions.
- ii) Iterative Multi-Scale Cross-Cue Mutual Disambiguation via Attention: We propose a multi-scale architecture that facilitates continuous bidirectional refinement between single-view semantic cues and multi-view geometric evidence. Through cross-attention and self-attention blocks, semantic context disambiguates noisy geometric matches, while geometric evidence sharpens semantic features. This establishes a mutual disambiguation loop that progressively resolves conflicts and enhances consistency, surpassing the recurrent refinement used in IterDepth and the simple concatenation-based fusion in existing methods.
- iii) Unified Closed-Loop Architecture for Joint Cost Purification and Cue Fusion: We move beyond the traditional aggregate-then-fuse pipeline by integrating cost-volume purification and cross-cue fusion into a unified closed-loop framework. This feedback cycle ensures that refined cost volumes improve fusion quality, while more discriminative fused features guide further aggregation, resulting in robust performance in visually degraded conditions. The proposed framework combines adaptive cost aggregation with interactive cue refinement, offering enhanced robustness in dynamic, low-texture, and occluded regions.

This paper establishes a synergistic cycle where improved cost aggregation enhances cross-cue fusion, and more consistent fused representations guide further cost refinement. This integration achieves competitive performance on multiple benchmarks, particularly in challenging regions with dynamic and low-texture scenes. The remainder of the paper is organized as follows: Section II reviews related work, Section III details the proposed methodology, Section IV presents

comprehensive experimental results and in-depth analysis, and Section V concludes the paper and outlines potential future directions.

2. Related work

This section provides a critical overview of the most pertinent developments in self-supervised single-frame and multi-frame monocular depth estimation methodologies, as well as the role of attention mechanisms in geometric learning and multi-cue information fusion. In addition, we connect the proposed method to the broader field of large-scale and cross-domain 3D reconstruction, highlighting how methods developed for terrestrial and underwater multi-view stereo address similar challenges in ill-conditioned visual environments, thereby situating the proposed framework within an interdisciplinary and application-oriented context.

2.1. Self-supervised single-frame monocular depth estimation

Single-frame monocular depth estimation aimed to predict a per-pixel depth map from a single RGB image. Traditional supervised approaches often relied on costly ground-truth annotations, such as LiDAR measurements (Fu et al., 2018) or manually labeled depth values (Chen et al., 2016). In contrast, self-supervised methods bypassed the need for such annotations by using photometric consistency across temporally adjacent video frames (Zhou et al., 2017) or stereo pairs (Garg et al., 2016; Godard et al., 2017), framing depth learning as a multi-view image-reconstruction task.

Recent advances in single-frame monocular depth estimation methods focused on addressing the inherent challenges through key directions including: more robust geometric and scale-aware losses (Gordon et al., 2019; Shu et al., 2020); discrete depth representations (Gonzalez & Kim, 2020; Johnston & Carneiro, 2020); feature-level reconstruction objectives (Zhan et al., 2018); weak supervision from automatically extracted sparse depth (Klodt & Vedaldi, 2018; Watson et al., 2019); specialized modules for handling occlusions and moving objects (Bian et al., 2021; Gordon et al., 2019; Klingner et al., 2020; Li et al., 2020; Yin & Shi, 2018; Zhang et al., 2024); and architecturally more expressive encoders and fusion designs (Guizilini et al., 2020), notable examples included SQLdepth (Wang et al., 2024), which constructed a self-cost volume from a single image for fine-grained geometry reasoning, MonoViT (Zhao et al., 2022), which introduced Transformers to compensate for convolutional locality, and WeatherDepth (Wang et al., 2024), which used curriculum contrastive learning to improve robustness under adverse weather. Further improvements were driven by the integration of transformer architectures and large-scale pre-training strategies: Metric3D (Yin & Chen, 2023) enhanced generalization by calibrating training data for camera and focal-length variations, DepthAnything (Yang et al., 2024) achieved robust zero-shot generalization via a large-scale data engine with automatic annotation and semantic priors from pre-trained encoders, and DepthCrafter (Hu et al., 2025) modeled depth estimation as a conditional probability distribution to boost temporal stability in video sequences.

Despite these advances, single-frame methods remained fundamentally limited by their reliance solely on appearance-based cues, which are inherently ambiguous in textureless regions, under occlusion, and in dynamic scenes where semantics conflict with geometry. Lacking explicit multi-view constraints, these approaches struggled to resolve depth ambiguities and are prone to systematic errors when monocular priors, such as object-size assumptions or contextual layout, failed to match the true 3D structure. Building on these insights, the proposed method integrated geometric priors, contextual information, and attention mechanisms into multi-frame learning, combining the scene-aware representational capacity of modern single-frame architectures with robust multi-view geometric reasoning to overcome appearance-based ambiguity while preserving semantic-structural advantages.

2.2. Self-supervised multi-frame monocular depth estimation

Multi-frame methods extended single-frame models by leveraging temporal information during inference, often achieving superior accuracy through explicit geometric matching. Early methods used RNNs to model temporal dependency (Patil et al., 2020; Wang et al., 2019; Zhang et al., 2019), but these were computationally costly and limited in geometric explicitness. The introduction of cost-volume-based frameworks (Watson et al., 2021) enabled direct encoding of multi-view correspondence, substantially improving robustness.

ManyDepth (Watson et al., 2021) introduced an adaptive cost-volume mechanism for variable-length inputs, with subsequent extensions integrating transformers for enhanced feature matching (Guizilini et al., 2022), iterative refinement networks for depth fusion (Feng et al., 2024), and occlusion (Feng et al., 2022) or motion-aware cost-volume designs (Dong et al., 2024; Miao et al., 2023; Wu et al., 2024) to handle dynamic content, while inference-time refinement methods further enabled online adaptation via photometric losses (Luo et al., 2020), albeit with increased computational cost. Furthermore, DualDepth (Feng et al., 2025) used a dual-path encoder with contrastive learning (decoupling single/multi-frame features for robust cross-condition performance) and a learnable offset interpolation module for efficient upsampling; Mono-ViFI (Liu et al., 2025) integrated optical-flow-based frame interpolation to align multi-frame features, achieving SOTA on KITTI and Cityscapes; Pose-Depth Joint Optimization (Shen et al., 2025) combined wide-baseline constraints and loop-closure consistency to reduce pose errors, plus uncertainty-guided depth sampling for better cost-volume matching. These approaches reflected a trend toward tighter integration of geometric priors, uncertainty modeling, and efficient upsampling.

Despite improvements in geometric consistency, current multi-frame methods still faced limitations due to insufficient use of geometric priors (e.g., surface normals, epipolar constraints) and weak coupling between cost-volume refinement and multi-cue fusion. As a result, they often failed to handle corrupted matching costs in low-texture, occluded, or dynamic regions, leaving local inconsistencies as a persistent challenge in depth estimation. In contrast, the proposed framework addressed these gaps through a unified architecture that jointly performed geometry-aware cost aggregation and iterative cross-cue attention, forming a closed-loop refinement process. By explicitly injecting geometric priors and context information into cost filtering and enabling multi-scale, attention-guided fusion between single-frame semantics and multi-frame geometry, the proposed approach not only rectified corrupted matching costs more effectively but also established a synergistic cycle: refined cost volumes enhance fusion, while improved fusion outcomes further guide aggregation. This design specifically enabled more consistent, reliable, and geometrically faithful depth estimation.

2.3. Deep multi-view stereo and large-scale reconstruction

Multi-frame monocular depth estimation was rooted in multi-view stereo (MVS), which reconstructed 3D scenes from multiple calibrated cameras. Early deep stereo networks (Kendall et al., 2017; Liang et al., 2018; Ummenhofer et al., 2017) introduced plane-sweep cost volumes and relied on ground-truth supervision, demonstrating that explicit geometric matching significantly boosted depth accuracy. Subsequent advances improved cost-volume regularization (Chang & Chen, 2018; Cheng et al., 2019; Wu et al., 2024; Zhang et al., 2020) and fused traditional stereo matching with monocular semantic cues (Facil et al., 2017; Martins et al., 2018). Supervised MVS systems (Hou et al., 2019; Im et al., 2019; Long et al., 2020) delivered impressive results but required dense depth and pose annotations, limiting their applicability in uncontrolled settings.

A more scalable direction was self-supervised MVS, which learned from photometric consistency without labeled ground-truth (Dai et al., 2019; Huang et al., 2021; Kar et al., 2017; Wang et al., 2019). Although

these methods alleviated annotation dependence, they remained sensitive to domain shifts, dynamic objects, and low-texture regions, which hindered deployment in real-world sequential data.

Beyond conventional stereo, recent advances in large-scale and cross-domain 3D reconstruction focused on handling extreme visual degradation in terrestrial, aerial, and underwater environments. For example, Yang et al. (2025a), (2025b) proposed robust geometric encoding and adaptive aggregation mechanisms to address challenges such as illumination variation, sparse textures, and environmental noise in city-scale and underwater scenes. These methods emphasized the critical role of geometric priors and context-aware matching under visually degraded conditions, which strongly aligned with the challenges encountered in monocular depth estimation for dynamic outdoor and adverse-weather scenarios.

The proposed framework aligned conceptually with these large-scale reconstruction paradigms by adapting their geometric-aware formulation, such as surface normal guidance and epipolar constraints, to the online monocular sequential setting. While methods like Yang et al. (2025a), (2025b) target offline, multi-view reconstruction, we integrated similar robustness strategies into a self-supervised, cost-volume-based pipeline that operated on sequential video data. This adaptation bridged insights from offline multi-view geometry to online monocular depth estimation, extending principled geometric reasoning to dynamic, texture-limited, and visually degraded real-world scenarios.

2.4. Attention mechanism for monocular depth estimation

Attention mechanisms were widely adopted to capture long-range dependencies in visual tasks (Vaswani et al., 2021). In depth estimation, self-attention and cross-attention were used to aggregate spatial context (Wu et al., 2024), enforce global coherence (Liu et al., Dec., 2023), and fuse semantic-geometric cues (Zhao et al., 2022). For example, Johnston and Carneiro (Johnston & Carneiro, 2020) applied self-attention to model global context among regions with similar disparity values, even when spatially discontinuous. Liu et al. (Liu et al., 2023) achieved fine-grained and globally consistent dense depth estimation using vision transformers as the backbone, while Zhao et al. (Zhao et al., 2022) introduced a cross-attention module to integrate implicit semantic cues into geometric representation. However, these methods often incurred high computational and memory costs. To alleviate this burden, several approaches introduced local or window-based attention mechanisms (Ruhkamp et al., 2021; Vaswani et al., 2021). Recent methods further integrated attention into multi-frame pipelines: DualDepth (Feng et al., 2025) used contrastive learning-based attention for adaptive feature weighting, Pose-Depth Joint Optimization (Shen et al., 2025) employed attention-guided fusion of geometric priors and multi-view cues, while our lightweight hybrid local attention balances expressive power and computational feasibility for iterative multi-scale cost volume refinement and cross-cue fusion.

In summary, the proposed framework unifies self-supervised monocular depth estimation and geometry-aware multi-frame reconstruction via an iterative design. Unlike prior methods that decouple cost aggregation from cue fusion or neglect geometric priors, it integrates such priors to enable cost purification, attention-guided cross-cue refinement and iterative multi-scale fusion. Leveraging large-scale MVS insights and tackling common challenges (texture sparsity, dynamic interference), this method enhanced the robustness of depth estimation for wider real-world and cross-domain deployment.

3. The proposed method

The proposed framework aims to address two core challenges in multi-frame depth estimation: (i) geometrically inconsistent and corrupted matching costs in dynamic or textureless regions, and (ii) ineffective fusion between single-frame (monocular) and multi-frame cues. To tackle these, we introduce two synergistic components built upon

distinct information pathways: Hybrid Geometric-Context Attention Module for cost volume rectification, and Iterative Multi-Scale Cross-Cue Fusion Module for robust integration of geometric and semantic information. The overall workflow establishes a mutual refinement loop between the geometry-dominant multi-frame path and the semantics-dominant single-frame path, is illustrated in Fig. 2 and summarized in the high-level pseudocode of Algorithm 1.

3.1. Overview and problem setup

Intuition and High-Level Design. The core idea in this paper is to create a mutual refinement loop between geometric matching evidence from multiple frames and semantic understanding from a single image. The process unfolds in three conceptual stages: i) Dual-Path Encoding, extract features from multi-frame inputs; construct an initial coarse (noisy) cost volume from multi-frame feature matching; ii) Mutual Refinement Loop, Iteratively refine the cost volume using a hybrid attention mechanism that integrates geometric smoothness and semantic context; iii) Cross-Cue Fusion, Fuse the rectified geometric features with semantic features through an iterative, attention-based architecture across multiple scales. This design ensures that ambiguous geometric matches are resolved using semantic context and geometric prior, while semantic predictions are geometrically regularized, leading to

Algorithm 1

Training Details.

Input: target frame I_t , source frames $I_s \in \{I_{t-N}, \dots, I_{t-1}, I_{t+1}, \dots, I_{t+N}\}$

Step 1: Feature Extraction
 $F_t = \text{Encoder}(I_t)$ // target frame features
 $F_s = \text{Encoder}(I_s)$ for each I_s // source frame features

Step 2: Depth Prior (Single-frame branch)
 $D_{\text{prior}} = \text{Decoder}_M(F_t)$ // initial depth estimation

Step 3: Pose Estimation
 $T_{t \rightarrow s} = {}^0\text{pose}(I_t, I_s)$ // relative pose

Step 4: Cost Volume Construction
 Initialize depth planes $P = \{d_i\}$ based on D_{prior}
 For each depth plane d_i :
 Warp F_s to target view using $T_{t \rightarrow s}$ and d_i
 Compute matching score with F_t
 $C_{\text{multi}} = \text{Aggregate matching scores over all source frames}$

Step 5: Hybrid Geometric and Context-aware Cost Aggregation
 (i) Geometric-consistent Aggregation
 Compute surface normal $\vec{n}$ from D_{prior}
 For each pixel (u, v) :
 Propagate adjacent costs using planar assumption and $\vec{n}$
 $C_{\text{geo}} = \text{GeometricAggregation}(C_{\text{multi}}, \vec{n})$
 (ii) Context-aware Aggregation
 $\mathbf{A}_x = \text{ComputeSpatialAttention}(F_t)$ // spatial attention weights
 $C_{\text{context}} = \text{ContextAggregation}(C_{\text{geo}}, \mathbf{A}_x)$

Step 6: Cross-cue Fusion Module
 $C_{\text{mono}} = \text{ConvertDepthToVolume}(D_{\text{prior}})$
 $F_{\text{multi}} = \text{Conv}_{1 \times 1}(C_{\text{context}})$
 $F_{\text{mono}} = \text{Conv}_{1 \times 1}(C_{\text{mono}})$
 Multi-cue Attention
 $F_{\text{multi_enh}} = \text{CrossAttention}(F_{\text{multi}}, F_{\text{mono}})$
 $F_{\text{mono_enh}} = \text{CrossAttention}(F_{\text{mono}}, F_{\text{multi}})$
 $F_{\text{fused}} = \text{Concat}(F_{\text{multi_enh}}, F_{\text{mono_enh}})$

Step 7: Channel Attention Fusion
 $F_{\text{channel_att}} = \text{ChannelAttention}(F_{\text{fused}}, F_t)$
 $F_{\text{final}} = F_{\text{channel_att}} + F_{\text{fused}}$

Step 8: Multi-scale Iterative Refinement
 For scale in $\{\text{coarse}, \text{medium}, \text{fine}\}$:
 For iteration in $[1, \dots, 4]$:
 Residual Update:
 $D_{\text{residual}} = \text{DepthDecoder}(F_{\text{final}}, F_t)$
 $D_t = D_t + 0.3 * D_t + D_{\text{residual}}$
 Pose Update:
 $T_{t \rightarrow s} = T_{t \rightarrow s} + \Delta T$

Step 9: Loss Computation (Training Phase)
 $L_{\text{ph}} = \text{PhotometricLoss}(I_t, I_s, D_t, T_{t \rightarrow s})$
 $L_{\text{smooth}} = \text{SmoothnessLoss}(D_t, I_t)$
 $L_{\text{total}} = L_{\text{ph}} + \lambda * L_{\text{smooth}}$

Output: High quality depth map D_t

more robust depth estimation.

Formal Problem Statement. Let I_t denote the target RGB image at time t . The goal is to predict its depth map D_t . In single-frame monocular depth estimation, the depth map D_t is predicted only from a single monocular RGB image I_t using an encoder-decoder network θ_{Depth} :

$$D_t = \theta_{\text{Depth}}(I_t). \quad (1)$$

In multi-frame settings, we additionally consider N preceding (and sometimes future) neighboring frames $\{I_{t-1}, \dots, I_{t-N}\}$ to exploit temporal consistency during training and testing. The multi-frame model is formulated as:

$$D_t = \theta_{\text{Depth}}(I_t, I_{t-1}, \dots, I_{t-N}). \quad (2)$$

The key difference is that self-supervised multi-frame depth estimation equips the single-frame encoder-decoder backbone with a cost volume construction module, one that leverages inter-frame geometric constraints to resolve scale ambiguity and improve accuracy. However, this volume tends to be corrupted in challenging scenarios. Existing methods typically handle cost volume filtering and depth cue fusion as two separate stages, whereas our key conceptual advance lies in tightly integrating these two processes into a synergistic loop.

For self-supervision during training, we utilize both preceding and subsequent frames without relying on ground-truth camera poses. For real-time inference, the proposed model adopts a causal test-time setting (only historical frames), unlike (Luo et al., 2020; Zhang et al., 2019). Following (Gordon et al., 2019), we assume camera intrinsics $\mathbf{K}$ are known, and relative poses $\{T_{t \rightarrow t-n}\}_{n=1}^N$ are estimated using a differentiable pose network θ_{pose} . The remainder of this paper elaborates on two core contributions: (i) the construction and rectification of the multi-view cost volume (Sec. III-B), and (ii) iterative fusion of this volume with single-view cues (Sec. III-C). Section III-D further summarizes the full set of training objectives.

3.2. Constructing the cost volume with adaptive sampling

We adopt a plane-sweep cost volume (Yao et al., 2018) to encode multi-view matching evidence across a range of depth values between the target frame I_t and its neighboring frames $I_{t \pm n}$, but enhance it with adaptive depth sampling guided by a monocular depth prior $\hat{d}_{\text{prior}}$. This ensures computational resources are focused on plausible depth ranges (Kendall et al., 2017). While this aligns with the depth-aware sampling in Long et al., (2022), which also highlights monocular cues for multi-view geometry, we extend the concept by tightly integrating the prior into sampling, matching, and fusion stages, forming a more unified geometry-semantics reasoning loop. The cost volume $C_{\text{multi}} \in \mathbb{R}^{H \times W \times N}$ is constructed in three adaptive steps.

i) Adaptive Depth Hypothesis Sampling

Instead of global uniform depth planes, we generate pixel-wise depth hypotheses $P = \{d_i\}_{i=1}^N$ within a local interval around $\hat{d}_{\text{prior}}(u, v)$.

$$d_i(u, v) = d_{\min}(u, v) + (d_{\max}(u, v) - d_{\min}(u, v)) \cdot i / N \quad (3)$$

where $d_{\min}(u, v) = \hat{d}_{\text{prior}}(u, v) \cdot r_{\min}$, $d_{\max}(u, v) = \hat{d}_{\text{prior}}(u, v) \cdot r_{\max}$. Here $r_{\min}$ and $r_{\max}$ are learned scaling factors. This data-dependent sampling focuses computation on high-probability depth ranges, boosting efficiency and discriminability.

ii) Feature Warping

For each hypothesis d_i and relative pose $T_{t \rightarrow t+n}$, we warp source frame features F_{t+n} to the target viewpoint via multi-view reprojection and compute the reprojection coordinate via homography $p_{t+n}' \sim D(p_t)^{-1} K T_{t+n \rightarrow t} K^{-1} D(p_t) p_t$. The source frame's multi-scale feature map F_{t+n} is then warped to the target viewpoint to yield $F_{t+n \rightarrow t, d_i}$.

iii) Matching Cost Calculation

The matching cost is constructed by computing pixel-wise similarity via normalized cross-correlation between warped $F_{t+n \rightarrow t, d_i}$ and the target

feature map F_t for each depth hypothesis. Averaging these scores across all source frames yields the final cost volume $C_{multi} \in \mathbb{R}^{H \times W \times N}$, where $C_{multi}(i, j, k) \in [0, 1]$ denotes the matching probability of spatial location (i, j) and depth hypothesis $k \in \{1, \dots, N\}$.

The depth-plane resolution should align with the feature map resolution used for cost volume construction, so that each depth plane corresponds accurately to a spatial location, thereby enabling precise geometric matching and semantic integration. Compared to methods sampling uniformly over a predefined inverse depth range, the prior-guided adaptive sampling enables data-dependent determination of depth planes for each pixel (u, v) . Following the practices of Li et al. (2023); Wu et al., (2024), the cost volume is iteratively aggregated and fused with multi-scale single-frame target features F_t to regress the accurate depth map D_t . As illustrated in Fig. 1, the proposed framework employs three scales to balance accuracy and computational efficiency. At each scale $s \in \{3, 2, 1\}$, the predicted depth map with spatial resolution $(H/2^s, W/2^s)$ is used as prior information to reduce the number of depth planes in higher-resolution regions.

3.3. Geometric and semantic consistent cost aggregation and fusion module

Conventional cost-volume-based monocular depth estimation models often generate inconsistent matching (e.g., erroneous holes in dynamic or texture-poor regions), owing to the absence of explicit semantic and geometric prior guidance. To resolve this issue, we propose the Hybrid Geometric-Contextual Aggregation (HGCA) module, which builds a bidirectional consistency loop between geometric priors and semantic context. By combining hybrid cost aggregation with cross-cue fusion, the module effectively rectifies corrupted matching costs in challenging scenarios and provides reliable multi-view geometric evidence for subsequent cross-cue fusion and depth estimation. As illustrated in Fig. 2, HGCA integrates a hybrid aggregation mechanism and a semantic-geometric fusion unit to robustly rectify corrupted matching costs in challenging regions. i) Geometric-Consistent Cost Aggregation

Global geometric consistency and physical plausibility require spatial aggregation across neighboring pixels, but naive aggregation neglects scene structure, leading to blurred object boundaries and error propagation. Inspired by the local planarity of real-world surfaces, we design a geometric-consistent cost aggregation module that incorporates local surface geometry as a prior to guide aggregation. This ensures cost fusion conforms to scene structure, enhancing cost volume discriminability and depth estimate geometric coherence.

The proposed method aligns the depth hypotheses of neighboring pixels from source views into a unified 3D coordinate system tied to the target frame. This alignment enables **geometry-aware cost propagation** based on a local planarity assumption. For a reference pixel (u_i, v_i) and its neighbor (u_j, v_j) , the corresponding 3D points $\mathbf{X}_i$ and $\mathbf{X}_j$ in the camera coordinate system are constrained by the estimated surface normal $\vec{n}$ via the planar consistency equation: $\vec{n}^T(\mathbf{X}(u_i, v_i) - \mathbf{X}(u_j, v_j)) = 0$. Under the local planar assumption, we establish a **depth transfer function** and relate depth hypotheses of neighboring pixels via a depth ratio r_{ji} derived from surface normals $\vec{n}$ by substituting the projection relation $\mathbf{X} = d(u, v) \cdot \begin{bmatrix} \frac{u - c_x}{f_x} & \frac{v - c_y}{f_y} & 1 \end{bmatrix}^T$:

$$r_{ji} = \frac{d(u_j, v_j)}{d(u_i, v_i)} = \frac{\vec{n}^T \begin{bmatrix} \frac{u_i - c_x}{f_x} & \frac{v_i - c_y}{f_y} & 1 \end{bmatrix}^T}{\vec{n}^T \begin{bmatrix} \frac{u_j - c_x}{f_x} & \frac{v_j - c_y}{f_y} & 1 \end{bmatrix}^T} \quad (4)$$

where (c_x, c_y) and (f_x, f_y) represent the camera principal point and focal lengths, respectively. The surface normal $\vec{n}$ is analytically computed from the current depth estimation using finite differences and iteratively refined in our alternating optimization loop.

Using the depth ratio r_{ji} , we can map a depth hypothesis d_i at pixel (u_i, v_i) to the corresponding hypothesis $d_j = r_{ji} \cdot d_i$ at pixel (u_j, v_j) , facilitating geometry-aware propagation of matching costs between neighboring pixels. For a depth hypothesis d_i at pixel (u_i, v_i) , we compute its propagated cost from neighbor (u_j, v_j) as:

$$C_{j \rightarrow i}(d_i) = C(u_j, v_j; r_{ji} \cdot d_i), \quad (5)$$

where $C(u_j, v_j; d_j)$ denote the matching cost at pixel (u_j, v_j) under depth hypothesis d_j , $C_{j \rightarrow i}(d_i)$ represents the contribution of neighboring pixel to the cost volume at (u_i, v_i) under depth d_i . The propagated matching cost $C_{j \rightarrow i}$ can be formally written as:

$$C_{j \rightarrow i}(d_i^0, \dots, d_i^L) = C_j(d_{i-j}^0, \dots, d_{i-j}^L) \quad (6)$$

In practice, for each candidate depth d_i at the reference pixel, we compute its propagated counterparts in the neighborhood, and aggregate their associated matching costs accordingly. Since d_j may not coincide exactly with one of the discretely sampled depth levels, we interpolate the matching cost $C(u_j, v_j; d_j)$ from the two nearest sampled

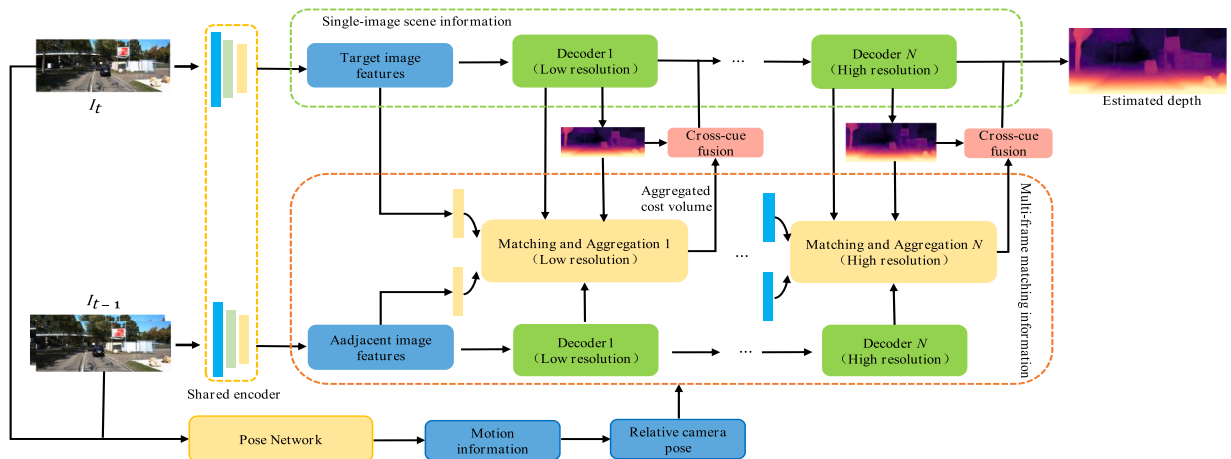


Fig. 1. An overview of the proposed overall framework, which consists of three main stages: feature extraction, matching and aggregation, multi-scale cross-cue iterative fusion. A shared encoder extracts features from both the target and source images. A multi-scale hybrid decoder then performs feature matching and cost aggregation, iteratively integrating scene information from a single image with multi-frame matching information from multiple images. The cost volume within the decoder is constructed using depth priors and the relative pose $[R, t]$ predicted in real time by the pose network.

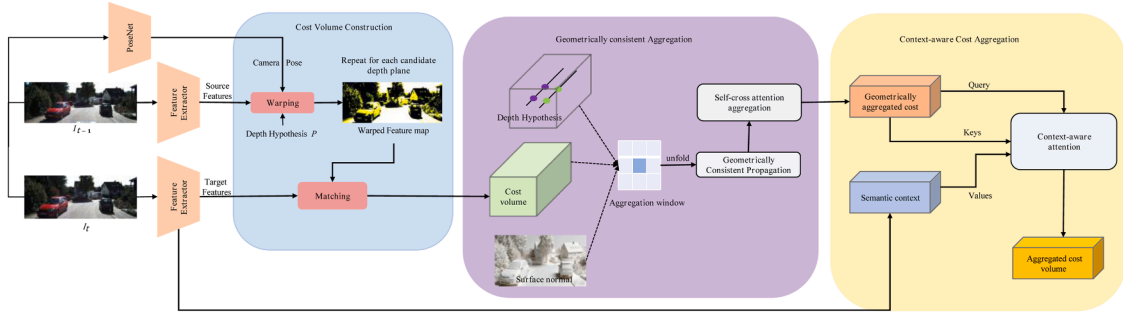


Fig. 2. Detailed architecture of the proposed geometrically and semantic consistent cost aggregation module. The module has two stages: Geometric Propagation enforces local planar consistency to reduce noise in low-texture and dynamic regions; Contextual Refinement uses channel-wise fusion and self-attention to enhance structural sharpness and coherence, especially around depth discontinuities.

depth values:

$$C(u_j, v_j; d_j) = \alpha \cdot C(u_j, v_j; d_j^{\downarrow}) + (1 - \alpha) \cdot C(u_j, v_j; d_j^{\uparrow}), \quad (7)$$

where $d_j^{\downarrow}$ and $d_j^{\uparrow}$ are the nearest discrete depth levels below and above d_j , respectively, and α is the interpolation weight determined by the relative position of d_j within the interval. This propagation is performed over a local window around each reference pixel, and the aggregated cost is obtained via a weighted combination of the original and propagated costs. Further mathematical details regarding this geometric formulation can be found in Wu et al. (2024).

The interpolated cost candidates are organized into a channel-wise tensor of size $H \times W \times (k \times k)$, where each channel corresponds to propagated costs from a neighboring location in a $k \times k$ window. This tensor undergoes channel-wise fusion via a $1 \times 1 \times k$ convolution, followed by a self-attention mechanism that captures non-local dependencies to enhance contextual consistency:

$$C_{geo} = \text{SelfAttention}(\text{Conv}_{1 \times 1 \times k}(C_{interp})), \quad (8)$$

where C_{interp} denotes the interpolated cost tensor and C_{geo} is the resulting geometrically consistent cost volume. This integrated process, combining depth-aware interpolation, local aggregation, and global attention, ensures that the cost volume respects local geometric constraints while preserving semantic coherence, thereby improving robustness in textureless regions and near depth boundaries. ii) Hybrid Geometric-Context Attention (HGCA) Module

To enhance matching cost reliability in regions where geometric cues are weak, such as textureless or highly dynamic areas, we introduce a Context-Aware Aggregation Module. This module builds on the observation that local appearance similarity often corresponds to consistent depth structure, even when geometric matching alone is ambiguous. As illustrated in Fig. 2, the module first learns pixel-wise affinity relationships within a local window using contextual appearance features, then uses these affinities to adaptively aggregate matching costs across neighboring positions. Formally, let F_t^c denote the context feature map extracted from the target image I_t . For a given pixel location $\mathbf{x}$, we derive a query vector $\mathbf{q}_{\mathbf{x}}$ from its local neighborhood using an identity-based query function $\phi_{\text{query}}(\cdot)$:

$$\mathbf{q}_{\mathbf{x}} = \phi_{\text{query}}(\mathbf{F}_t^c(N_{\mathbf{x}})), \quad (9)$$

where $N_{\mathbf{x}}$ represents a local window centered at $\mathbf{x}$.

Simultaneously, key features $\mathbf{k}_{\mathbf{x}'}$ for all neighboring positions $\mathbf{x}'$ within an enlarged or shifted window around $\mathbf{x}$ are extracted via a lightweight 3×3 convolutional operator $\phi_{\text{keys}}(\cdot)$:

$$\mathbf{k}_{\mathbf{x}} = \phi_{\text{keys}}(\mathbf{F}_t^c(N'_{\mathbf{x}})). \quad (10)$$

The query and key features are then concatenated along the channel dimension and fed into a small convolutional network $\phi_{\text{spatial}}(\cdot)$, which consists of two successive 1×1 convolutional layers followed by a

Softmax normalization. This process yields a spatially varying attention weight map $\mathbf{A}_{\mathbf{x}}$ that encodes the affinity between $\mathbf{x}$ and its surrounding pixels:

$$\mathbf{A}_{\mathbf{x}} = \text{Softmax}(\phi_{\text{spatial}}(\phi_{\text{query}}(\mathbf{F}_t^c(\mathbf{x})), \phi_{\text{keys}}(\mathbf{x}, \mathbf{F}_t^c))) \quad (11)$$

The pixel-wise attention weights $\mathbf{A}_{\mathbf{x}}$ reflect how strongly each neighboring pixel should contribute to the refined matching cost at location $\mathbf{x}$, based on their contextual similarity. These weights are used to perform a weighted aggregation of the geometrically consistent cost volume C_{geo} within the local window:

$$C_{\text{context}}(\mathbf{x}) = \mathbf{x}' \in N_{\mathbf{x}} \sum \mathbf{A}_{\mathbf{x}}(\mathbf{x}') \cdot C_{geo}(\mathbf{x}'), \quad (12)$$

where C_{context} denotes the contextually refined cost volume. Finally, the values are obtained through a projection function $\phi_{\text{values}}(\cdot)$ implemented as a 1×1 convolution. The matching costs are then further aggregated using the normalized attention weights within the shifted window. This adaptive aggregation combines geometric plausibility with appearance similarity, preserving sharp boundaries while improving robustness in challenging areas. iii) Cross-Cue Semantic-Geometric Fusion

To effectively integrate the complementary strengths of single-frame scene priors and multi-frame geometric matching, we propose a Multi-Geometric Attention Mechanism that facilitates a bidirectional, attention-guided interaction between the rectified multi-frame cost volume C_{multi} and a monocular depth volume $C_{\text{mono}} \in \{0, 1\}^{H \times W \times N}$ (obtained by discretizing the monocular depth map D_t). Unlike independent or late-fusion strategies, this mechanism enables each cue to refine the other based on structural and contextual relationships, promoting deeper integration.

The monocular depth map D_t is encoded as a Monocular Depth Volume $C_{\text{mono}} \in \{0, 1\}^{H \times W \times N}$, where each pixel is represented as a one-hot vector across N depth planes to match the dimensions of the multi-frame cost volume C_{multi} . This binary volume preserves the absolute scale and structural prior from the single-image pathway, and is subsequently refined using the same context-aware aggregation module to enhance its spatial consistency.

Given the multi-scale cost volumes $C_{\text{multi}}, C_{\text{mono}} \in \mathbb{R}^{H \times W \times N}$, we first apply shallow convolutional layers to obtain down-sampled depth features $F_{\text{multi}}, F_{\text{mono}} \in \mathbb{R}^{H \times W \times M}$, where M denotes the channel dimension after projection. These features are then passed through Multi-cue **Cross Attention** module, which enhances each depth cue by leveraging the structural relationships present in the other cue. As illustrated in Fig. 3, the two feature streams interact in parallel through a cross **attention mechanism**. Taking the enhancement of multi-frame features via monocular cues as an example. **Feature Transformation:** The monocular feature F_{mono} is transformed into **query** Q_{mono} and **key** K_{mono} matrices via learned linear projections, while the multi-frame feature F_{multi} is mapped to a **value** matrix V_{multi} through a convolutional operation $f(\cdot, \theta)$. **Relation Computation:** All three matrices are reshaped to size (HW, M) to enable matrix operations. The attention weights are

computed by measuring the similarity between queries from the monocular stream and keys from the same stream, followed by Softmax normalization:

$$\mathbf{A}_{\text{mono} \rightarrow \text{multi}} = \text{Softmax}(\mathbf{Q}_{\text{mono}} \mathbf{K}_{\text{mono}}^T). \quad (13)$$

These weights capture the **intra-structural relationships** within the monocular depth prior, such as object layout and relative depth ordering, which are semantically rich but geometrically ambiguous. **Feature Enhancement:** The attention weights are then applied to the multi-frame value features to produce the enhanced representation:

$$\mathbf{F}_{\text{multi-enh}} = \text{Softmax}(\mathbf{Q}_{\text{mono}} \otimes \mathbf{K}_{\text{mono}}^T) \otimes \mathbf{V}_{\text{multi}} \quad (14)$$

This step effectively **infuses semantic structural guidance** from the monocular stream into the multi-frame geometric representation, helping to resolve matching ambiguities in textureless or occluded regions. Symmetrically, the monocular features are refined through attention computed from the multi-frame stream:

$$\mathbf{F}_{\text{multi-enh}} = \text{Softmax}(\mathbf{Q}_{\text{multi}} \otimes \mathbf{K}_{\text{multi}}^T) \otimes \mathbf{V}_{\text{mono}} \quad (15)$$

where $\mathbf{Q}_{\text{multi}}$ and $\mathbf{K}_{\text{multi}}$ are derived from $\mathbf{F}_{\text{multi}}$, and $\mathbf{V}_{\text{mono}}$ from $\mathbf{F}_{\text{mono}}$. This allows **geometric consistency** from multi-view matching to regularize the monocular depth prior, sharpening boundaries and suppressing outliers.

The two enhanced features, $\mathbf{F}_{\text{multi-enh}}$ and $\mathbf{F}_{\text{mono-enh}}$, are then combined, e.g., through summation or concatenation followed by convolution, to form a **jointly refined cost volume** that embodies both geometrically consistent matching evidence and semantically informed structural priors. This tightly coupled, attention-mediated fusion ensures that the complementary strengths of both cues are preserved and mutually reinforced, leading to more robust and accurate depth estimation.

The enhanced features $\mathbf{F}_{\text{multi-enh}}$ and $\mathbf{F}_{\text{mono-enh}}$ are concatenated to form the fused feature $\mathbf{F}_{\text{fused}}$. To preserve fine-grained information from the original cues, we also compute $\mathbf{F}_{\text{cat}} = \text{Cat}(\text{Conv}(\mathbf{C}_{\text{multi}}), \text{Conv}(\mathbf{C}_{\text{mono}}))$ and add it as a residual connection to $\mathbf{F}_{\text{fused}}$, yielding the aggregated cross-cue feature $\mathbf{F}$. The fusion module further incorporates contextual information via soft channel attention. First, an element-wise sum of $\mathbf{F}$ and the context feature $\mathbf{F}_c^i$ is globally average-pooled to obtain a descriptor $\mathbf{F}_a^i$. A lightweight network $\phi_{\text{channel}}(\cdot)$, with two 1×1 convolutions and a Softmax function, then computes channel attention weights $\mathbf{A}_{ch} = \phi_{\text{channel}}(\mathbf{F}_a^i)$. The final fused feature is obtained as:

$$\mathbf{F}_{\text{im-fused}} = \mathbf{A}_{ch} \mathbf{F}_c^i + (1 - \mathbf{A}_{ch}) \mathbf{F} \quad (16)$$

$\mathbf{F}_{\text{im-fused}}$ is then passed to the decoder along with image context features to predict the final depth map $D_t \in \mathbb{R}^{H \times W}$.

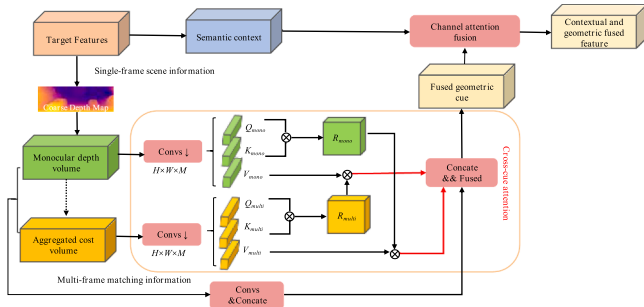


Fig. 3. Overall framework for the cross-attention and channel attention-based multi-cue fusion module. The module fuses multi-frame cost volumes and monocular depth cues using bidirectional cross-attention, then enhances features with channel attention to provide a unified, coherent basis for depth estimation.

3.4. Optimization and iterative training framework

The proposed training framework is designed to synergistically leverage single-frame scene priors and multi-frame geometric matching, building upon established paradigms in self-supervised monocular (Godard et al., 2019) and multi-frame (Watson et al., 2021; Wu et al., 2024) depth estimation. The core supervisory signal is derived from photometric consistency across adjacent video frames, without requiring ground-truth depth.

Alternately Iterative Optimization. Following the philosophy of recent multi-frame methods (Long et al., 2022; Watson et al., 2021), we construct a feature-metric plane-sweep cost volume to encode multi-view matching evidence. Inspired by iterative refinement architectures like CAT-Net (Wu et al., 2024), our pipeline employs a dual-branch alternating optimization scheme (Fig. 2), where a disparity head $f_d(\cdot)$ and a pose head $f_p(\cdot, \cdot)$ are updated cyclically. Both heads adopt network structures similar to DRO (Gu et al., 2023).

At each iteration i and for each scale, the disparity map is refined using a proportionate update strategy, with the update operation at each iteration performed as follows: $D_t^i = D_t^{i-1} + 0.3 D_t^{i-1} \rho_t^i$, where D_t^i and D_t^{i-1} are the disparity maps of the i th and $(i-1)$ th iterations respectively, with $\rho_t^i \in [-1, 1]$ being the residual proportion predicted by the disparity head, and $\alpha = 0.3$ is a scaling factor controlling the update magnitude. This multiplicative form allows adaptive adjustment relative to the current disparity estimate.

The pose network, meanwhile, is updated following a conventional additive strategy: $T_{t \rightarrow t+n}^i = T_{t \rightarrow t+n}^{i-1} + \Delta T_{t \rightarrow t+n}^i$, where $T_{t \rightarrow t+n}^i$ and $T_{t \rightarrow t+n}^{i-1}$ are the estimated relative poses of the i th and $(i-1)$ th iterations respectively, and $\Delta T_{t \rightarrow t+n}^i$ is the pose residual predicted by the pose network. This alternating process enables joint refinement of geometry and motion, where improved depth aids pose estimation and vice versa. This alternating process enables joint refinement of geometry and motion, where improved depth aids pose estimation and vice versa.

The primary training signal comes from photometric reprojection. The depth network θ_{depth} and the pose network θ_{pose} are trained alternately using a consistent loss function. The self-supervision signal is derived from the photometric reprojection error between the target image and its synthesized version, which is generated from the source image. Using the current estimates, depth D_t from the depth network and relative camera pose $T_{t \rightarrow t+n}$ from the pose network, the scene is synthesized from the viewpoint of I_t by warping pixels from neighboring source frames I_{t+n} , where $n \in \{-1, 1\}$. The synthesized image serving as the counterpart to I_t is:

$$I_{t+n \rightarrow t} = I_{t+n} < \text{proj}(D_t, T_{t \rightarrow t+n}, K) > \quad (17)$$

Here, $\langle \cdot \rangle$ denotes the bilinear sampling operator, and proj computes the 2D coordinates corresponding to the depths in D_t when projected onto the image plane of I_{t+n} . For each pixel, the photometric loss L_{ph} is minimized by selecting the best-matching source image, achieved by taking the per-pixel minimum over the reconstruction-based photometric error pe ,

$$L_{ph} = \min_n [pe(I_t, I_{n \rightarrow t})] \quad (18)$$

The photometric error pe is defined as a convex combination of Structural Similarity (SSIM) and the L1 norm between the original and reconstructed images. This loss is minimized over all pixels across four different output scales during training. Additionally, an edge-aware smoothness loss $L_s = |\partial_x D_t^*| e^{-|\partial_x I_t|} + |\partial_y D_t^*| e^{-|\partial_y I_t|}$, adopted from Monodepth2 (Godard et al., 2019), is applied using mean-normalized inverse depth $D_t^* = D_0 / \bar{D}_0$ to prevent underestimation of depth values. Further details can be found in Godard et al. (2019). The overall loss function in the self-supervised training pipeline is thus a combination of the photometric loss and the smoothness loss:

$$L_{self-sup} = L_{ph} + \lambda L_s. \quad (19)$$

where λ balances the two terms. This loss is computed and back-propagated across four output scales to enforce multi-scale consistency. The iterative optimization is applied in a coarse-to-fine manner: at each scale, initial coarse depth maps are generated and refined through several iterations in the bottom-up phase. Then, features and depth estimates from coarser scales guide the refinement at finer scales in the top-down phase, ensuring global consistency and local detail recovery. This hierarchical strategy progressively reduces ambiguity and stabilizes training, ultimately yielding a depth map that is both globally coherent and locally precise.

4. Experiments

In this section, we conducted extensive experiments to validate the effectiveness of the proposed self-supervised multi-frame monocular depth estimation framework, which integrated geometric-context-aware attention for cost aggregation and iterative multi-scale cross-cue fusion. We began by presenting the implementation details and dataset descriptions. To comprehensively evaluate the performance and generalization capability of the proposed method, we adopted a two-fold evaluation strategy. First, we conducted quantitative and qualitative evaluations across multiple domains by training and testing the proposed model on three distinct scenarios: autonomous driving (using the KITTI dataset (Geiger et al., 2012)), indoor scenes (using the NYUv2 dataset (Kohli et al., 2012)), and adverse weather conditions (using the Weather-KITTI dataset (Wang et al., 2024)). Subsequently, we performed a zero-shot cross-dataset evaluation on Cityscapes (Cordts et al., 2016) (outdoor driving), ScanNet (Dai et al., 2017) (indoor), and DENSE (Bijelic et al., 2020) (adverse weather) to rigorously assess its generalization ability. Finally, we carried out detailed ablation studies with both quantitative metrics and visual comparisons to assess the contribution of each proposed component.

4.1. Implementation details and datasets

During training, self-supervision signal was derived from frames $\{I_{t-1}, I_b, I_{t+1}\}$, while at test time we fixed $N = 1$, constructing the cost volume from frames $\{I_b, I_{t-1}\}$. The proposed architecture was implemented in PyTorch 1.13.1 and trained on an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. The single-frame pathway in the proposed architecture built upon Monodepth2 (Godard et al., 2019), using an encoder in θ_{depth} composed of the first five layers of ResNet18 (He et al., 2016). The pose network θ_{pose} and depth decoder followed the design of (Gu et al., 2023), establishing the DRO method (Gu et al., 2023) as our baseline. Coarse depth maps at each scale were iteratively trained for 15 epochs in a bottom-up manner. The full network was then trained with the AdamW optimizer (Kingma & Adam, 2014) for 50 epochs, using a batch size of 6. The initial learning rate was set to 10^{-4} , and reduced by a factor of 10 after 30 epochs, following a cosine annealing schedule. Gradient clipping with a maximum norm of 1.0 was applied to stabilize training. The number of recurrent iterations was empirically set to 4. In line with common practice (Godard et al., 2019; Watson et al., 2021), the feature encoder modules were initialized with weights pre-trained on ImageNet (Russakovsky et al., 2015). Depth predictions were clipped at 80 m according to KITTI evaluation guidelines, and all metrics were computed within this range, incorporating a center crop following (Godard et al., 2019). Based on empirical values, hyperparameters $[r_{min}, r_{max}]$ for scales $\{3, 2, 1\}$ were set to $\{[0.5, 1.5], [0.6, 1.4], [0.8, 1.2]\}$ respectively (Long et al., 2022), with $\lambda = 0.001$. Unless specified otherwise, all models were trained at resolutions of 480×160 for KITTI dataset and WeatherKITTI dataset, and 384×288 for NYUv2 dataset. We applied standard data augmentations, including random color jittering and horizontal flipping, to input images for both depth and pose networks, following the practice in Monodepth2 (Godard et al., 2019).

The proposed model was primarily designed and evaluated on two widely used outdoor autonomous driving datasets, which featured dynamic traffic participants, low-texture road surfaces, and complex occlusion patterns typical of real-world driving scenarios. To further assess its robustness and generalizability, we conducted extensive experiments under diverse challenging conditions (including indoor environments and extreme weather) on multiple representative challenging datasets. All evaluations strictly followed the established dataset-specific protocols to ensure fair and consistent comparisons with existing methods.

i) Outdoor Autonomous Driving Datasets.

The KITTI dataset (Geiger et al., 2012) was preprocessed by removing static frames following the protocols of existing methods (Eigen et al., 2014; Zhou et al., 2017), resulting in 39,810 triplets for training and 4424 for validation and testing using the Eigen splits (Eigen et al., 2014; Godard et al., 2019). While this split was widely used in single-frame depth estimation, it was employed in multi-frame settings recently (Wang et al., 2019). The KITTI Eigen testing set included the first frame of 22 video sequences, which lacked a previous frame. These images were still included in our evaluation; however, since no previous frame was available, depth predictions for these cases were derived solely from the current frame. To further ensure the reliability of evaluations in dynamic and occluded regions, we also adopt the improved ground truth provided by Uhrig et al. (2017), which offered denser and more accurate depth annotations. Following the standard evaluation protocol for autonomous driving scenes, we employed the metrics established by Eigen et al. (Eigen et al., 2014), including absolute relative error (Abs Rel), squared relative error (Sq Rel), root mean squared error (RMSE), RMSE in log space (RMSElog), and accuracy under thresholds $\delta_1, \delta_2, \delta_3$ (where $\delta_1 < 1.25^1$), consistent with (Long et al., 2022; Wu et al., 2024).

Cityscapes Dataset (Cordts et al., 2016). To evaluate the cross-domain generalization capability of the proposed method in autonomous driving scenarios, we directly tested the KITTI pre-trained models on the Cityscapes dataset (Cordts et al., 2016). Following the protocol in Zhou et al. (2017), we used 1525 test frames for evaluation. Consistent with the setting for KITTI, the predicted depths were clipped at 80 m, and evaluations were performed only on ground-truth depths within this range. ii) Indoor Scene Datasets (NYUv2 & ScanNet):

To evaluate generalization and cross-domain transferability across varying depth ranges, illumination conditions, and structural layouts, we further conducted experiments on two standard indoor datasets. A key challenge for indoor depth estimation was the prevalence of low-texture surfaces (e.g., uniform walls, floors, and furnishings), which offered limited local visual cues for conventional matching-based methods.

NYU Depth v2 (Kohli et al., 2012). We conducted both training and validation of the proposed framework using the NYU Depth V2 dataset, a widely-used indoor benchmark with depth values up to 10 m, containing scenes with varied lighting and cluttered layouts. Comprising 582 indoor scenes captured by a Microsoft Kinect camera, this dataset encompassed a wide range of everyday environments, including office spaces, residential living areas, and kitchen settings. To align with the experimental protocols established in prior literature (Guo et al., 2024), we adopted the same training/testing partition: the training subset comprises 21,483 images drawn from 283 distinct scenes. All training images were undistorted and sampled at 10-frame intervals to preserve temporal variation, and we leveraged sequences of three successive frames $\{I_{t-1}, I_b, I_{t+1}\}$ during training to enable effective multi-frame feature extraction. Model evaluation was carried out on the official test split, which includes 654 annotated target frames I_t and their corresponding reference frames I_{t-1} .

ScanNet (Dai et al., 2017) was a large-scale indoor RGB-D dataset comprising 1513 densely annotated 3D reconstructions of indoor environments, captured with handheld RGB-D sensors. Each scene spanned diverse indoor spaces, such as offices, living rooms, and kitchens, featuring challenging visual characteristics including severe occlusions,

texture-less surfaces, and complex geometric layouts. The dataset provided per-frame depth maps, camera poses, and instance-level semantic annotations, making it a widely adopted benchmark for 3D scene understanding. To assess the zero-shot cross-scene generalization capability of the proposed depth model, which was only trained exclusively on NYUv2 dataset, we evaluated on its official test split following the established protocol in Li et al., Feb. (2023). Input images were set to a resolution of 320×240 . This setup allowed us to systematically examine how well the proposed method generalizes to unseen indoor scenes, particularly when the training data was limited to a single structured indoor domain. iii) Adverse Weather Dataset:

The WeatherKITTI Dataset (Wang et al., 2024) was built upon the original KITTI dataset and integrated six distinct weather augmentations (rain, fog, snow, etc.). It offered more realistic simulations of complex real-world driving scenarios than other synthetic weather datasets, making it well-suited for training models under adverse weather conditions. This dataset adopted the same train/validation/test splits as KITTI. For training, we adopted Zhou's split (Zhou et al., 2017), which includes 39,810 training images and 4424 validation images. For testing, we used the Eigen split (Eigen et al., 2014), comprising 697 test images, ensuring consistency with established evaluation protocols.

To further validate the generalization performance of the proposed depth estimation method under challenging real-world adverse weather conditions, we additionally utilized the DENSE dataset (Bijelic et al., 2020). This benchmark features driving scenes captured in a diverse range of extreme weather (rain, snow, fog, etc.), which were characterized by severe illumination variations, degraded visibility, and realistic sensor noise. Boasting around 1500 annotated snowy scenes, DENSE outperformed counterparts such as CADC in annotation quality and scene coverage, while containing fewer artifacts induced by dynamic objects and occlusions. For consistent evaluations and fair comparisons with prior methods (Wang et al., 2024), we uniformly sampled one frame every three from the dataset to construct a curated test set of 500 images. Its LiDAR-based depth annotations were refined via the DROR algorithm to filter out snowflake-induced noise, thus enhancing the reliability of ground-truth data. All images were cropped to a resolution of 1880×924 to alleviate sensor-induced distortion.

4.2. Monocular depth estimation results

We first evaluated the proposed method on the KITTI dataset (Geiger et al., 2012) using two standard data splits (Eigen et al., 2014; Uhlig et al., 2017) to demonstrate its effectiveness in monocular depth estimation. i) Quantitative Results:

KITTI Eigen Split: We first compared the proposed method with state-of-the-art self-supervised single- and multi-frame monocular depth estimation methods that also leverage cost volumes on the KITTI Eigen split dataset (Eigen et al., 2014). As shown in Table 1, the proposed approach achieved higher accuracy in both AbsRel and SqRel across multiple resolutions than prior single- and multi-frame methods. Furthermore, the proposed method consistently outperformed previous approaches across most metrics, especially in SqRel and RMSE, which indicated fewer large errors in dynamic and low-texture regions. It also surpassed the closely related CAT-Net (Wu et al., 2024) across all metrics. These results demonstrated that the proposed method effectively enhanced monocular depth estimation accuracy by enforcing local geometric and semantic consistency aggregation within the cost volume, and adopting a more effective multi-scale fusion strategy between single-frame and multi-frame cues, all of which reduced large-error regions.

KITTI Improved Ground-Truth: To more reliably evaluate model performance in dynamic and occluded regions, where the original Eigen split (Eigen et al., 2014) provided sparse annotations, we further assessed the proposed approach using the improved ground truth from Uhlig et al. (2017), which delivered denser and more accurate depth labels through multi-frame aggregation. As demonstrated in Table 2 on

this enhanced benchmark (Uhlig et al., 2017), the proposed method consistently outperformed previous approaches across most metrics and resolutions. These results not only confirmed the robustness of the proposed method in handling occluded regions and moving objects, but also further validate the efficacy of the proposed modules. Moreover, they underscored the significant potential of multi-frame self-supervised approaches in addressing complex environmental challenges inherent in real-world driving scenarios.

Cityscapes Generalization: To further evaluate the cross-dataset generalization capability of the proposed method, we trained the proposed model on the KITTI dataset (Geiger et al., 2012) and directly tested it on the Cityscapes dataset (Cordts et al., 2016). Quantitative cross-dataset results on Cityscapes were presented in Table 3. For fair comparison, we also evaluated SC-Depth (Bian et al., 2021), Monodepth2 (Godard et al., 2019), and ManyDepth (Watson et al., 2021) using their publicly available pre-trained models. The proposed approach achieved better performance across most metrics, particularly in Sq Rel and RMSE, which demonstrated the effectiveness of the proposed aggregation and fusion strategy. Without fine-tuning, the proposed method maintained strong cross-dataset performance, highlighting its superior domain adaptation capability, consistent with prior observations (Watson et al., 2021; Wu et al., 2024) that multi-frame methods generally outperformed single-frame approaches in cross-domain generalization, as they jointly utilized geometric and appearance representations. Further supporting the finding that geometric cues exhibited better domain adaptability than appearance-based features, these results validated that the proposed model's geometric-aware fusion module significantly enhanced its zero-shot generalization ability on Cityscapes.

Indoor NYUv2 dataset evaluation: We evaluated the proposed method on the official NYUv2 test set against state-of-the-art supervised and self-supervised depth estimation approaches. As presented in Table 4, the proposed multi-frame model achieved consistent and competitive performance across all metrics compared to single-frame and multi-frame baselines, especially for self-supervised indoor methods, verifying the effectiveness of our multi-frame design for indoor depth estimation. These results demonstrated that the proposed multi-frame method had strong adaptability to the structural characteristics of indoor scenes, also enabling it to achieve state-of-the-art results for self-supervised monocular depth estimation indoors.

Generalization Evaluation on Indoor ScanNet Dataset. To further validate the indoor generalization capability of the proposed method, we conducted zero-shot evaluation on the unseen ScanNet dataset without any retraining, and the results were summarized in Table 5. The proposed method achieved favorable performance on ScanNet; despite the inevitable performance decline caused by domain shift, the degradation was relatively slight. This demonstrated that the proposed method had good generalization ability to unseen indoor scenes with significant differences, and exhibited strong robustness to variations in depth range and structural disparities. Table 6

Adverse Weather Evaluation on WeatherKITTI Datasets: To assess broader applicability, quantitative results on the WeatherKITTI dataset in Table 6 showed that the proposed method still could produce plausible values across all evaluation metrics under general adverse conditions. However, in scenarios involving severe weather or extreme illumination changes, such as intense darkness or brightness, its performance remained limited compared to methods specifically designed with adaptation mechanisms for such environments. This gap stemmed primarily from the lack of effective style adaptation and well-designed contrastive loss functions, which were critical under highly variable visual conditions.

Generalization Evaluation on the DENSE Dataset. To further assess the broader and practical applicability of our method, we conduct zero-shot evaluations of models pre-trained on WeatherKITTI using the real-world DENSE dataset, which encompasses challenging adverse-weather scenarios characterized by illumination changes and reduced visibility.

Table 1

QUANTITATIVE Evaluation RESULTS for Monocular DEPTH ESTIMATION ON THE KITTI EIGEN SPLIT DATASET (Geiger et al., 2012; Eigen et al., 2014). WE COMPARED RESULTS ACROSS LOW AND HIGH RESOLUTIONS WITH AND WITHOUT ADDITIONAL SUPERVISION. “INPUT FRAMES” REFERRED TO THE NUMBER OF FRAMES USED AS INPUT DURING TESTING. THE BEST RESULTS WERE HIGHLIGHTED IN BOLD. “M” INDICATES MONOCULAR VIDEO SUPERVISION, AND “SEM” DENOTED THE USE OF SEMANTIC SEGMENTATION INFORMATION.

Methods	Input frames	Train	Resolution	Error metric				Accuracy metric		
				Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
SfMLearner (Zhou et al., 2017)	1	M	640 × 192	0.148	1.344	5.927	0.247	0.803	0.922	0.964
PackNet-SfM (Guizilini et al., 2020)	1	M	640 × 192	0.111	0.785	4.601	0.189	0.878	0.960	0.982
Monodepth2 (Godard et al., 2019)	1	M	640 × 192	0.115	0.903	4.863	0.193	0.877	0.959	0.981
Bian et al. (Bian et al., 2021)	1	M	832 × 256	0.114	0.813	4.706	0.191	0.873	0.960	0.982
SGDDepth (Klingner et al., 2020)	1	M+Sem	640 × 192	0.113	0.835	4.693	0.191	0.879	0.961	0.981
FixUnDepth (Chen et al., 2022)	1	M	416 × 128	0.129	0.976	4.908	0.203	0.848	0.951	0.979
RM-Depth (Hui, 2022)	1	M	640 × 192	0.108	0.710	4.513	0.183	0.884	0.964	0.983
Zhang et al. (Zhang et al., 2024)	1	M	640 × 192	0.105	0.786	4.572	0.182	0.890	0.964	0.983
Lee et al. (Lee et al., 2021)	2(−1, 0)	M	640 × 192	0.114	0.876	4.715	0.191	0.872	0.955	0.981
ManyDepth (Watson et al., 2021)	2(−1, 0)	M	640 × 192	0.098	0.770	4.459	0.176	0.900	0.965	0.983
DynamicDepth (Feng et al., 2022)	2(−1, 0)	M+Sem	640 × 192	0.096	0.720	4.458	0.175	0.897	0.964	0.984
DepthFormer (Guizilini et al., 2022)	2(−1, 0)	M	640 × 192	0.090	0.661	4.149	0.175	0.905	0.967	0.984
DRO (Gu et al., 2023)	2(−1, 0)	M	480 × 160	0.099	0.813	4.478	0.192	0.881	0.957	0.980
CAT-Net (Wu et al., 2024)	2(−1, 0)	M	480 × 160	0.093	0.718	4.463	0.180	0.899	0.963	0.985
MOVEDepth (Wang et al., 2023)	2(−1, 0)	M	640 × 192	0.089	0.663	4.216	0.169	0.904	0.966	0.984
IterDepth (Feng et al., 2024)	2(−1, 0)	M	640 × 192	0.093	0.723	4.317	0.174	0.908	0.965	0.984
MonoViFi (Liu et al., 2025)	3(−1,0,1)	M	640 × 192	0.091	0.589	4.088	0.166	0.912	0.969	0.984
ProDepth (Woo et al., 2024)	2(−1, 0)	M	640 × 192	0.086	0.629	4.139	0.166	0.918	0.969	0.984
Ours	2(−1, 0)	M	480 × 160	0.085	0.621	4.124	0.165	0.919	0.971	0.985
Monodepth2 (Godard et al., 2019)	1	M	1024 × 320	0.115	0.882	4.701	0.190	0.879	0.961	0.982
PackNet-SfM (Guizilini et al., 2020)	1	M	1280 × 384	0.107	0.802	4.538	0.186	0.889	0.962	0.981
Liu et al. (Liu et al., 2023)	1	M	1280 × 320	0.106	0.673	4.379	0.180	0.886	0.965	0.984
Shu et al. (Shu et al., 2020)	1	M	1024 × 320	0.104	0.729	4.481	0.179	0.893	0.965	0.984
Zhang et al. (Zhang et al., 2024)	1	M	1024 × 320	0.100	0.712	4.369	0.176	0.901	0.967	0.984
Wang et al. (Wang et al., 2020)	2(−1, 0)	M	1024 × 320	0.106	0.773	4.491	0.185	0.890	0.962	0.982
ManyDepth (Watson et al., 2021)	2(−1, 0)	M	1024 × 320	0.093	0.715	4.245	0.172	0.909	0.966	0.983
DRO (Gu et al., May 2023)	2(−1, 0)	M	960 × 320	0.088	0.797	4.464	0.212	0.899	0.959	0.980
CAT-Net (Wu et al., June 2024)	2(−1, 0)	M	960 × 320	0.086	0.681	4.246	0.180	0.902	0.960	0.983
Ours	2(−1, 0)	M	960 × 320	0.081	0.612	4.102	0.162	0.920	0.972	0.986

Table 2

Quantitative Evaluation at Different Input Resolutions on the KITTI Dataset with Improved Ground Truth (Uhrig et al., 2017).

THE COLUMN “INPUT FRAMES” REFERRED TO THE NUMBER OF INPUT FRAMES USED DURING TESTING. THE BEST RESULTS IN EACH CATEGORY WERE HIGHLIGHTED IN BOLD.

Methods	Input frames	Train	Resolution	Error metric				Accuracy metric		
				Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
SfMLearner (Zhou et al., 2017)	1	M	640 × 192	0.176	1.532	6.129	0.244	0.758	0.921	0.971
Vid2Depth (Mahjourian et al., 2018)	1	M	640 × 192	0.134	0.983	5.501	0.203	0.827	0.944	0.981
GeoNet (Yin & Shi, 2018)	1	M	640 × 192	0.132	0.994	5.240	0.193	0.833	0.953	0.985
EPC++ (Luo et al., Oct., 2020)	1	M	832 × 256	0.120	0.789	4.755	0.177	0.856	0.961	0.987
Monodepth2 (Godard et al., 2019)	1	M	640 × 192	0.090	0.545	3.942	0.137	0.914	0.983	0.995
Johnston et al. (Johnston & Carneiro, 2020)	1	M	640 × 192	0.081	0.484	3.716	0.126	0.927	0.985	0.996
PackNet-SfM (Guizilini et al., 2020)	1	M	640 × 192	0.078	0.420	3.485	0.121	0.931	0.986	0.996
Patil et al. (Patil et al., 2020)	N	M	640 × 192	0.087	0.495	3.775	0.133	0.917	0.983	0.995
Wang et al. (Wang et al., 2020)	2(−1, 0)	M	640 × 192	0.082	0.462	3.739	0.127	0.923	0.984	0.996
ManyDepth (Watson et al., 2021)	2(−1, 0)	M	640 × 192	0.070	0.399	3.455	0.113	0.941	0.989	0.997
Long et al. (Long et al., 2022)	2(−1, 0)	M	640 × 192	0.068	0.366	3.338	0.110	0.946	0.989	0.997
DynamicDepth (Feng et al., 2022)	2(−1, 0)	M+Sem	640 × 192	0.068	0.362	3.454	0.111	0.943	0.991	0.998
DRO (Gu et al., May 2023)	2(−1, 0)	M	480 × 160	0.070	0.388	3.568	0.117	0.933	0.986	0.996
CAT-Net (Wu et al., June 2024)	2(−1, 0)	M	480 × 160	0.064	0.354	3.368	0.109	0.944	0.988	0.997
Ours	2(−1, 0)	M	480 × 160	0.061	0.349	3.354	0.108	0.946	0.989	0.998
Monodepth2 (Godard et al., 2019)	1	M	1024 × 320	0.091	0.531	3.742	0.135	0.916	0.984	0.995
ManyDepth (Watson et al., 2021)	2(−1, 0)	M	1024 × 320	0.064	0.345	3.116	0.103	0.949	0.989	0.997
DRO (Gu et al., May 2023)	2(−1, 0)	M	960 × 320	0.057	0.342	3.201	0.123	0.952	0.989	0.996
CAT-Net (Wu et al., June 2024)	2(−1, 0)	M	960 × 320	0.054	0.271	2.827	0.096	0.958	0.990	0.997
Ours	2(−1, 0)	M	960 × 320	0.052	0.265	2.674	0.093	0.960	0.991	0.998

As shown in Table 7, the proposed method retains reasonable generalization capability under these out-of-distribution and perceptually degraded conditions. Although performance experiences a domain-shift-induced degradation compared to models explicitly trained for such extreme conditions, our approach still demonstrates plausible and robust depth estimation in scenarios with significant illumination variations and limited visibility, confirming its practical utility in real-world autonomous driving environments.

2) Qualitative Comparisons

Outdoor Scenes (KITTI & Cityscapes): Fig. 4 qualitatively compared the proposed method with recent self-supervised approaches on the KITTI Eigen split. While ManyDepth (Watson et al., 2021) and CAT-Net (Wu et al., 2024) (first two columns) struggled to estimate depth accurately for moving vehicles, the proposed method (last column) produced notably sharper and more precise predictions, particularly for thin structures like signs and pedestrians. This demonstrated its improved ability to handle moving objects. Moreover, although some predicted depth maps appeared visually similar to those from existing models, the

Table 3

Generalization Performance of Monocular Depth Estimation on the Cityscapes Dataset. All models were trained exclusively on the KITTI dataset and evaluated directly on Cityscapes. Best results were highlighted in Bold.

Methods	Error metric				Accuracy $\delta < 1.25$
	Abs Rel	Sq Rel	RMSE	RMSE log	
SC-Depth (Bian et al., 2021)	0.177	2.117	9.780	0.267	0.720
Monodepth2 (Godard et al., 2019)	0.173	1.883	8.968	0.241	0.757
ManyDepth (Watson et al., 2021)	0.170	1.790	8.362	0.236	0.748
DRO (Gu et al., 2023)	0.160	1.792	8.804	0.256	0.758
CAT-Net (Wu et al., 2024)	0.153	1.699	8.339	0.232	0.772
Ours	0.139	1.479	8.001	0.215	0.782

corresponding error metrics confirmed a meaningful reduction in large-magnitude errors. Fig. 5 presented qualitative comparisons on Cityscapes (Cordts et al., 2016), showcasing results across diverse urban scenes, including streets, highways, and intersections. The proposed method effectively captured fine structural features such as building facades, road boundaries, and pedestrian paths, while generalizing robustly to challenging conditions including motion blur, illumination variations, and occlusion, maintaining reliable performance on both large objects (e.g., trucks) and small structures (e.g., road signs). These results illustrated typical success cases (e.g., improved handling of moving objects and low-texture regions) and representative failure modes (e.g., errors in highly occluded or reflective areas).

Figs. 6 and 7 presented qualitative results of the proposed method on indoor datasets. Specifically, Fig. 6 showed in-distribution performance on the NYUv2 test set, with training and evaluation conducted in the same domain. Fig. 7 illustrated zero-shot generalization results on ScanNet, where the model, trained exclusively on NYUv2 and without any ScanNet-specific fine-tuning, successfully preserved overall room layouts and recovers fine-grained object boundaries. This compelling visual outcome clearly demonstrated the model's strong cross-domain transfer capability for indoor depth estimation.

4.3. Ablation studies

In this section, we conducted detailed ablation experiments to verify the contribution of each proposed component and to justify the design of the geometric-semantic aggregation and iterative multi-scale fusion strategy. As shown in Table 8, the proposed method introduced two key innovations: (i) a Hybrid Geometric-Context Attention Module for cost volume rectification, and (ii) an Iterative Multi-Scale Cross-Cue Fusion Module for robust integration of monocular and multi-frame cues. To evaluate their individual impacts, we compared incremental model

variants built upon the DRO baseline (Gu et al., 2023).

Effect of Depth-Prior-Guided Cost Volume. First, we incorporated a monocular depth prior to guide the sampling of depth hypotheses in the cost volume construction. This yielded measurable gains across all error metrics (e.g., Abs Rel improved from 0.099 to 0.094), confirming that embedding semantic priors into the geometric matching space enhanced both efficiency and discriminability.

Separate Analysis of Geometric and Contextual Aggregation. To isolate their individual contributions, we first added only the Context-aware Cost Aggregation module, which leveraged appearance similarity to smooth the cost volume. Then, we added only the Geometric-based Cost Aggregation module, which enforced local planar constraints to propagate reliable matching evidence. Both modules independently improved over the prior-guided baseline, demonstrating that geometric and contextual cues provided complementary refinements.

Effect of Hybrid Cost Volume Aggregation. Next, we integrated the proposed Hybrid Geometric-Context Attention (HGCA) module, which enforced local geometric consistency while leveraging semantic context to rectify noisy matching costs. This led to a further reduction in Abs Rel to 0.090 and a notable improvement in accuracy metrics ($\delta < 1.25$ increases to 0.896), demonstrating the module's effectiveness in handling dynamic and textureless regions.

Table 5

PRESENTS ZERO-SHOT GENERALIZATION RESULTS ON THE SCANNet DATASET (Dai et al., 2017). HERE, SELF-SUP. REFERRED TO MODELS TRAINED IN A SELF-SUPERVISED MANNER USING ONLY IMAGE SEQUENCES, AND THE POSE. LABEL INDICATED THAT COLMAP CAMERA POSES WERE INCORPORATED DURING TRAINING.

Methods	Supervision	Abs Rel↓	RMSE↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
TrainFlow (Zhao et al., 2020)	Self-Sup.	0.189	0.415	0.726	0.927	0.980
SC-Depth (Bian et al., Sep., 2021)	Self-Sup.	0.169	0.392	0.749	0.938	0.983
Monodepth2 (Godard et al., 2019)	Self-Sup.	0.170	0.401	0.730	0.948	0.991
Manydepth (Watson et al., 2021)	Self-Sup.	0.154	0.373	0.779	0.951	0.988
F ² Depth (Guo et al., 2024)	Self-Sup.	0.125	0.310	0.845	0.969	0.992
SIM-MultiDepth (Guo et al., 2024)	Self-Sup.	0.118	0.297	0.862	0.974	0.994
ManyDepth (Watson et al., 2021)	Self-Sup.	0.121	0.301	0.858	0.969	0.991
Ours	Self-Sup.	0.117	0.295	0.861	0.976	0.994

Table 4

MONOCULAR DEPTH EVALUATION ON NYUv2. SUP: ✓ - SUPERVISED LEARNING USING GROUND-TRUTH DEPTH, AND ✗ - NOT USING GROUND-TRUTH DEPTH. WE ACHIEVED THE BEST RESULTS AMONG ALL SELF-SUPERVISED METHODS ON NYUv2 EVEN OUTPERFORM SOME SUPERVISED METHODS.

Methods	Sup	Train on NYUv2	AbsRel↓	RMSE↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
PlaneReg (Yu et al., 2019)	✓	✓	0.134	0.503	82.7	96.3	99.0
DORN (Fu et al., 2018)	✓	✓	0.115	0.509	82.8	86.5	99.2
P3Depth (Patil et al., 2022)	✓	✓	0.104	0.356	89.8	98.1	99.6
DDP (Chen et al., 2-3 Oct., 2023)	✓	✓	0.094	0.329	92.1	99.0	99.8
TranFlow (Zhao et al., 2020)	✗	✓	0.189	0.686	70.1	91.2	97.8
Zhang (Zhang et al., 2022)	✗	✓	0.177	0.634	73.3	93.6	-
Bian et al. (Bian et al., Sep., 2021)	✗	✓	0.157	0.593	78.0	94.0	98.4
Monodepth2 (Godard et al., 2019)	✗	✓	0.170	0.617	74.8	94.2	98.6
ADPDepth (Song et al., 2023)	✗	✓	0.165	0.592	75.3	93.4	98.1
F ² Depth (Guo et al., 2024)	✗	✓	0.158	0.583	77.9	94.7	98.7
MonoIndoor++ (Li et al., 2023)	✗	✓	0.132	0.517	83.4	96.1	99.0
SIM-MultiDepth (Guo et al., 2024)	✗	✓	0.152	0.567	79.2	95.0	98.8
ManyDepth (Watson et al., 2021)	✗	✓	0.154	0.571	78.9	94.6	98.1
Ours	✗	✓	0.128	0.494	83.7	96.4	98.9

Table 6

Quantitative Results on WeatherKITTI Datasets.

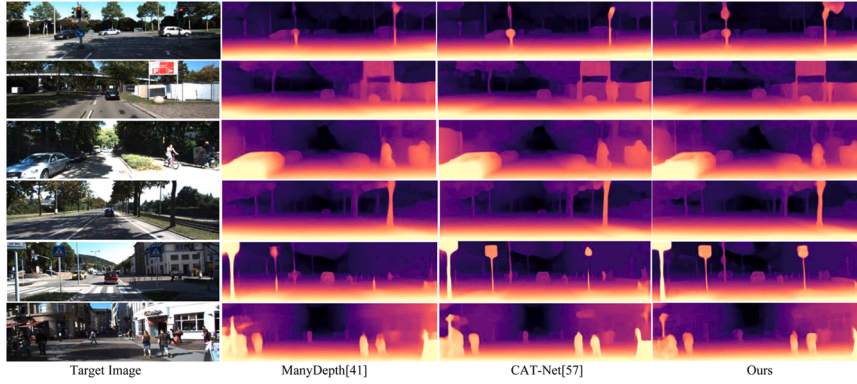
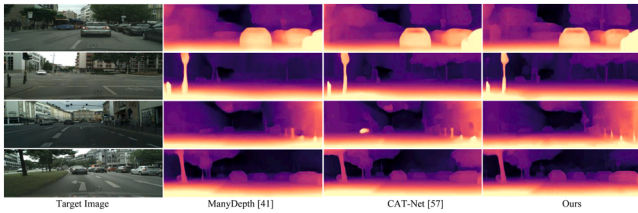
ERROR-BASED METRICS (LOWER VALUES INDICATE BETTER PERFORMANCE) AND ACCURACY-BASED METRICS (HIGHER VALUES INDICATE BETTER PERFORMANCE) ARE REPORTED.

Methods	Abs Rel↓	SqRel↓	RMSE↓	RMSElog↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
MonoViT (Zhao et al., Sep., 2022)	0.120	0.899	5.111	0.200	0.857	0.953	0.980
Robust-Depth (Saunders et al., 2023)	0.107	0.791	4.604	0.183	0.883	0.963	0.983
Robust-Depth* (Saunders et al., 2023)	0.103	0.806	4.517	0.180	0.894	0.964	0.983
WeatherDepth (Wang et al., 2024)	0.103	0.738	4.414	0.178	0.892	0.965	0.984
DRO (Gu et al., May 2023)	0.134	0.965	5.224	0.214	0.826	0.942	0.971
CAT-Net (Wu et al., 2024)	0.133	0.957	5.196	0.209	0.829	0.946	0.978
Ours	0.119	0.892	5.012	0.198	0.858	0.955	0.981

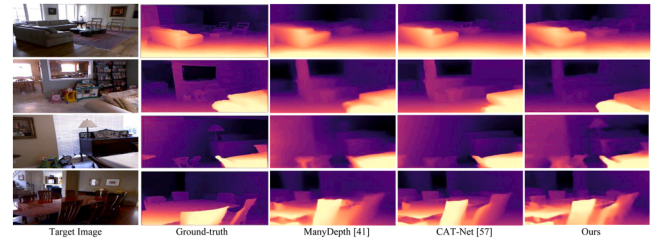
Table 7

Zero-shot Evaluation is Conducted on the Dense Snowy Dataset, where Models are Tested without any Additional Fine-tuning on this Specific scene.

Method	Abs Rel↓	SqRel↓	RMSE↓	RMSElog↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
MonoViT (Zhao et al., Sep., 2022)	0.162	2.063	9.787	0.262	0.762	0.904	0.960
Robust-Depth (Saunders et al., 2023)	0.157	1.992	8.945	0.240	0.786	0.923	0.971
Robust-Depth* (Saunders et al., 2023)	0.157	2.050	8.951	0.243	0.785	0.921	0.969
WeatherDepth (Wang et al., 2024)	0.157	2.000	9.021	0.243	0.781	0.919	0.968
DRO (Gu et al., May 2023)	0.174	2.176	10.126	0.283	0.732	0.868	0.944
CAT-Net (Wu et al., June 2024)	0.172	2.168	10.012	0.272	0.745	0.888	0.951
Ours	0.161	2.069	9.695	0.259	0.764	0.905	0.961

**Fig. 4.** Qualitative results on the KITTI dataset (Geiger et al., 2012) demonstrated that the proposed method produced more accurate boundaries and finer-grained details in challenging areas such as pedestrians, road signs, and moving vehicles compared to other multi-frame methods including ManyDepth (Watson et al., 2021) and CAT-Net (Wu et al., June 2024).**Fig. 5.** Qualitative cross-dataset generalization results on Cityscapes (Cordts et al., 2016). All models are trained on KITTI (Geiger et al., 2012) and evaluated zero-shot.

Effect of Cross-cue Attention Fusion. Introducing the Cross-cue Attention Fusion module, which enabled bidirectional interaction between monocular and multi-frame features via mutual attention, brought significant improvement (Abs Rel: 0.088, $\delta < 1.25$: 0.908). This demonstrated the importance of deep, attention-driven fusion over

**Fig. 6.** Qualitative evaluation on the NYUv2 dataset (Kohli et al., 2012) compared to other multi-frame methods including ManyDepth (Watson et al., 2021) and CAT-Net (Wu et al., June 2024).

simple feature concatenation.

Effect of Channel Attention Fusion. Adding the Channel Attention Fusion mechanism, which performed soft channel-wise selection

Table 8

ABLATION STUDIES OF THE PROPOSED METHOD FOR MONOCULAR DEPTH ESTIMATION ON THE KITTI EIGEN SPLITS DATASETS (Eigen et al., 2014). BEST RESULTS WERE HIGHLIGHTED IN BOLD.

Methods	Error metric				Accuracy metric		
	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
DRO (Baseline) (Gu et al., May 2023)	0.099	0.813	4.478	0.192	0.881	0.957	0.980
+Depth prior-based Cost Volume	0.094	0.807	4.411	0.187	0.883	0.958	0.981
+ only Context-aware Cost Aggregation	0.093	0.805	4.405	0.185	0.886	0.958	0.981
+ only Geometric-based Cost Aggregation	0.092	0.803	4.397	0.183	0.889	0.959	0.981
+Hybrid Cost Volume Aggregation	0.090	0.793	4.365	0.181	0.896	0.960	0.982
+ Cross-cue Attention Fusion	0.088	0.787	4.245	0.176	0.908	0.965	0.983
+Channel Attention Fusion	0.086	0.766	4.218	0.172	0.911	0.967	0.983
+ Iterative Multi-scale Feature Fusion	0.085	0.621	4.124	0.165	0.919	0.971	0.985

between context features and aggregated multi-cue representations, further reduced errors (Abs Rel: 0.086, $\delta < 1.25$: 0.911). This highlighted the benefit of adaptive feature recalibration at the channel level.

Effect of Iterative Multi-Scale Feature Fusion. Finally, the full model with iterative multi-scale refinement achieved the best results across all metrics (Abs Rel: 0.085, $\delta < 1.25$: 0.919), underscoring the synergy of our coupled design. The progressive refinement loop allowed geometry and semantics to mutually correct and enhanced each other, leading to robust and accurate depth estimation even in challenging scenarios.

In summary, the ablation study clearly demonstrated that each proposed component contributed positively, and their progressive integration formed a coherent and effective framework for multi-frame depth estimation. The results also confirmed the synergistic effect of combining geometric, contextual, and attention-based fusion mechanisms in a unified pipeline.

Table 9 presented quantitative results with varying numbers of iterations. When the number of iterations is zero, the initial depth map relies solely on the single-frame prior. Performance improved gradually as the number of iterations increases, but both error and accuracy metrics saturated when exceeding four iterations. Thus, four iterations were selected as the default setting in the proposed method.

Table 10 presented sensitivity analysis on sampling range hyperparameters. To balance model performance and computational complexity, we conducted a sensitivity analysis on the sampling range hyperparameters $[r_{min}, r_{max}]$ used for depth hypothesis sampling at different scales. These parameters directly influenced the coverage and granularity of the depth hypotheses in the cost volume, thereby affecting both estimation accuracy and computational overhead. This configuration set in our paper ensured that the depth hypotheses were sufficiently dense to capture fine-grained geometric details while avoiding excessive computational cost.

To complement the quantitative ablation analysis, we presented qualitative visualizations in Fig. 8. As shown in Fig. 8 (a), we visualized the iterative refinement of depth predictions across different iteration steps (1st, 2nd, 4th, and 6th iterations). The depth maps gradually improved in boundary precision and detail recovery, and the results stabilized after 4 iterations, which aligned with the earlier finding that performance saturated beyond this point. In Fig. 8(b), we conducted a qualitative ablation study by removing key components from the full framework. Removing the Hybrid Geometric-Consistent Aggregation module resulted in blurred edges and inconsistent depth predictions in low-texture regions. The absence of the Multi-scale Cross-cue Fusion module led to noticeable fusion artifacts in dynamic regions and reduced sharpness at depth boundaries, indicating the importance of bidirectional information exchange between geometric and semantic features. Furthermore, a comparison among the single-frame baseline,

Table 9

PERFORMANCE EVALUATION OF MONOCULAR DEPTH ESTIMATION ON THE KITTI EIGEN SPLIT (Geiger et al., 2012) WITH VARYING NUMBERS OF ITERATIONS.

Iter.	Error metric				
	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$
1	0.122	0.923±0.02	4.868±0.01	0.188	0.865
2	0.099	0.754±0.01	4.639±0.02	0.186	0.889
3	0.091	0.709±0.03	4.262±0.02	0.174	0.904
4	0.085	0.621±0.01	4.124±0.01	0.165	0.919
5	0.084	0.621±0.02	4.124±0.03	0.165	0.919
6	0.085	0.621±0.01	4.123±0.01	0.165	0.918

and our multi-frame full fusion method clearly showed that the hybrid fusion strategy recovers finer details and effectively suppressed outliers in moving regions.

4.4. Efficiency analysis

Table 11 compared model size, runtime, and FLOPs with recent multi-frame methods. The proposed model maintained the inference speed while improving accuracy, offering a practical balance for deployment. In terms of model complexity, as summarized in Table X, the proposed method incurred only a moderate increase in memory consumption compared to CAT-Net (Gu et al., 2023). Furthermore, the proposed method remained comparable to DRO in both model size and computational complexity. The efficiency of the proposed model stemmed primarily from the use of lightweight local attention and a purely convolutional network architecture, which together maintained competitive performance without substantially increasing resource demands.

Table 10

Sensitivity analysis of sampling range hyperparameters on the KITTI eigen split.

Scale	$[r_{min}, r_{max}]$	Abs Rel ↓	Sq Rel ↓	RMSE ↓	$\delta < 1.25$ ↑
3	[0.3, 1.7]	0.087	0.635	4.158	0.915
3	[0.5, 1.5]	0.085	0.621	4.124	0.919
3	[0.7, 1.3]	0.086	0.628	4.139	0.917
2	[0.4, 1.6]	0.086	0.630	4.145	0.916
2	[0.6, 1.4]	0.085	0.621	4.124	0.919
2	[0.8, 1.2]	0.087	0.633	4.162	0.914
1	[0.6, 1.4]	0.086	0.629	4.141	0.917
1	[0.8, 1.2]	0.085	0.621	4.124	0.919
1	[1.0, 1.0]	0.089	0.645	4.197	0.915

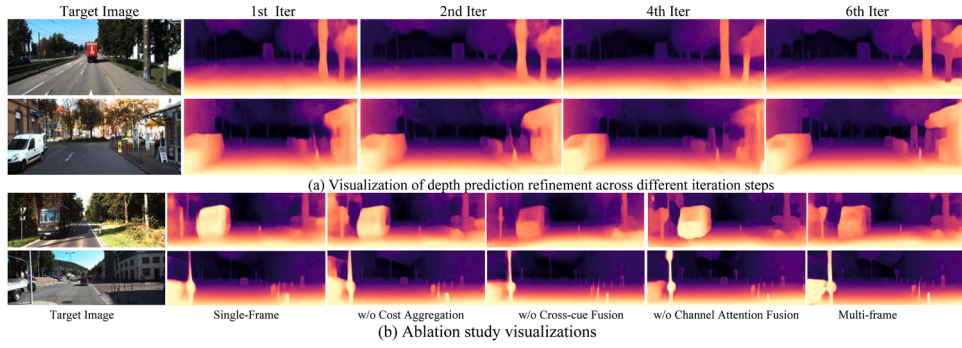


Fig. 8. Qualitative results of iterative refinement and ablation studies on KITTI Eigen splits dataset (Geiger et al., 2012). Qualitative ablation studies visualized by omitting individual modules from the whole framework.

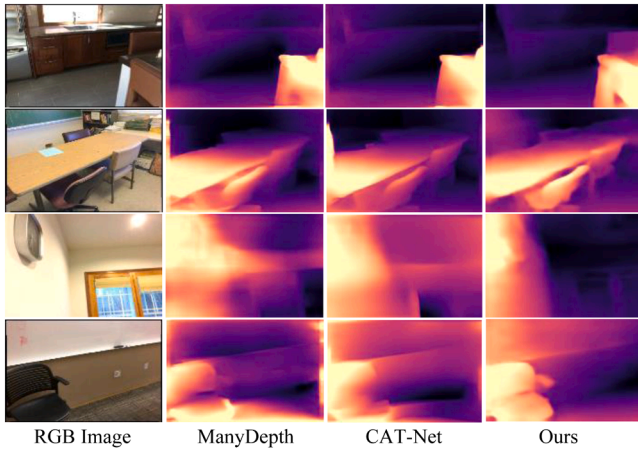


Fig. 7. Visualization of cross-dataset generalization on ScanNet (Dai et al., 2017). Compared with ManyDepth (Watson et al., 2021) and CAT-Net (Wu et al., June 2024), the proposed method achieved more accurate and fine-grained depth estimation in unseen scenes.

4.5. Limitations & future work

While the proposed method advanced multi-frame depth estimation, it still struggled with extremely fast-moving objects and severe weather artifacts (e.g., heavy snow). Future work will integrate explicit motion estimation and weather-invariant representation learning.

5. Conclusions

In this paper, we present a self-supervised multi-frame monocular depth estimation framework designed to address matching inconsistencies in dynamic scenes. The core of the method lies in a hybrid geometric-semantic aggregation module that iteratively refines corrupted cost volumes, enabling robust depth inference under challenging conditions such as texture-sparse regions, occlusion, and object motion. By integrating surface normal constraints, contextual cues, and attention-based feature fusion, the proposed approach ensures both local geometric consistency and semantic reliability. Extensive experiments on outdoor, indoor, and adverse-weather benchmarks demonstrate that the proposed method not only achieves high accuracy but also

Table 11

Quantitative analysis of model complexity: parameters, computational cost (Mult-Adds), and memory footprint (INPUT SIZE: 480×160 , BATCH SIZE: 1).

Methods	Params	Mult-Adds	Memory	FLOPs (G)	Time (s)
DRO (Baseline) (Gu et al., May 2023)	12,680,775	19.48 G	328.81 MB	715.4	0.11
CAT-Net	12,692,115	19.54 G	332.92 MB	726.3	0.12
Ours	12,698,483	19.54 G	332.96 MB	728.1	0.14

exhibits strong generalization across domains. This validates the effectiveness of combining geometric priors with semantic aggregation to better leverage multi-view information while retaining the flexibility of monocular cues. Beyond depth estimation for autonomous driving, the proposed framework offers a generalizable strategy for geometry-aware feature fusion under weak supervision. Its robustness to visual degradation makes it suitable for practical applications such as large-scale 3D reconstruction from unordered images, robotic navigation in complex settings, and remote sensing. Future work will focus on incorporating explicit motion modeling to further improve performance in highly dynamic scenarios, and extending the framework to cross-domain tasks including indoor mapping and aerial imagery analysis.

CRediT authorship contribution statement

Jiaojiao Fang: Writing – original draft, Methodology.

Declaration of competing interest

We hereby declare that there are no known conflicts of interest associated with this publication, and there has been no significant financial support for this work that could have influenced its outcome.

References

- Bian, J.-W., et al. (2021). Unsupervised scale-consistent depth learning from video. *International Journal of Computer Vision*, 129(9), 2548–2564.
- Bijelic, M., Gruber, T., Mannan, F., Kraus, F., Ritter, W., Dietmayer, K., & Heide, F. (2020). Seeing through fog without Seeing fog: Deep multimodal sensor fusion in unseen adverse weather. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*.
- Chang, J.-R., & Chen, Y.-S. (2018). Pyramid stereo matching network. In *CVPR*.

- J. Chen, Z. Xie, H. Liu, X. Liu, L. Tu, L. Li, and P. Luo, "DDP: Diffusion model for dense visual prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Paris, France, 2023, pp. 21741–21752.
- Chen, S., Pu, Z., Fan, X., & Zou, B. (2022). Fixing defect of photometric loss for self-supervised monocular depth estimation. *IEEE Transactions on Circuits and System for Video Technology*, 32(3), 1328–1338.
- Chen, W., Fu, Z., Yang, D., & Deng, J. (2016). Single-image depth perception in the wild. *In NeurIPS*.
- Cheng, X., Wang, P., & Yang, R. (2019). Learning depth with convolutional spatial propagation network. *PAMI*.
- Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Franke, U., Roth, S., & Schiele, B. (2016). The cityscapes dataset for semantic urban scene understanding. *In CVPR*.
- Dai, A., Chang, A. X., Savva, M., Halber, M., Funkhouser, T., & Nießner, M. (2017). Scannet: Richly-annotated 3d reconstructions of indoor scenes. *In Proc. IEEE Conf. Comput. Vis. Pattern Recognit* (pp. 5828–5839).
- Dai, Y., Zhu, Z., Rao, Z., & Li, B. (2019). MVS2: Deep unsupervised multi-view stereo with multi-view symmetry. *In 3DV*.
- Dong, Y.-J., Zhang, F.-L., & Zhang, S.-H. (2024). Mal: Motion-aware loss with temporal and distillation hints for self-supervised depth estimation. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 7318–7324). IEEE.
- Eigen, D., Puhrsch, C., & Fergus, R. (2014). Depth map prediction from a single image using a multi-scale deep network. *In NeurIPS*.
- Facil, J. M., Concha, A., Montesano, L., & Civera, J. (2017). Single-view and multi-view depth fusion. *IEEE Robotics and Automation Letters*.
- Feng, C., Chen, Z., Zhang, C., Hu, W., Li, B., & Lu, F. (2024). Iterdepth: Iterative residual refinement for outdoor self-supervised multi-frame monocular depth estimation. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(1), 329–341.
- Feng, C., Zhang, C., Chen, Z., Hu, W., Lu, K., & Ge, L. (2025). Self-supervised monocular depth estimation with dual-path encoders and offset field interpolation. *IEEE Transactions On Image Processing : A Publication of the IEEE Signal Processing Society*, 34.
- Feng, Z., Yang, L., Jing, L., Wang, H., Tian, Y., & Li, B. (2022). Disentangling object motion and occlusion for unsupervised multi-frame monocular depth. *In Proc. Eur. Conf. Comput. Vis* (pp. 228–244).
- Fu, H., Gong, M., Wang, C., Batmanghelich, K., & Tao, D. (2018). Deep ordinal regression network for monocular depth estimation. *In CVPR*.
- Garg, R., BG, V. K., & Reid, I. (2016). Unsupervised CNN for single view depth estimation: Geometry to the rescue. *In ECCV*.
- Geiger, A., Lenz, P., & Urtasun, R. (2012). Are we ready for autonomous driving? The KITTI Vision benchmark suite. *In CVPR*.
- Godard, C., Mac Aodha, O., & Brostow, G. J. (2017). Unsupervised monocular depth estimation with left-right consistency. *In CVPR*.
- Godard, C., Mac Aodha, O., Firman, M., & Brostow, G. J. (2019). Digging into self-supervised monocular depth estimation. *In ICCV*.
- Gonzalez, J. L., & Kim, M. (2020). Forget about the LiDAR: Self-supervised depth estimators with MED probability volumes. *NeurIPS*.
- Gordon, A., Li, H., Jonschkowski, R., & Angelova, A. (2019). Depth from videos in the wild: Unsupervised monocular depth learning from unknown cameras. *In ICCV*.
- Gu, X., et al. (2023). DRO: Deep recurrent optimizer for video to depth. *IEEE Robotics and Automation Letters*, 8(5), 2844–2851.
- Guizilini, V., Ambrus, R., Chen, D., Zakharov, S., & Gaidon, A. (2022). Multi-frame self-supervised depth with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 160–170).
- Guizilini, V., Ambrus, R., & Pillai, S. (2020). Allan Raventos, and Adrien Gaidon. 3D packing for self-supervised monocular depth estimation. *In CVPR*.
- Guo, X., Zhao, H., Shao, S., et al. (2024a). SIM-MultiDepth: Self-supervised indoor monocular multi-frame depth estimation based on texture-aware masking. *Remote Sensing*, 16(12).
- Guo, X., Zhao, H., Shao, S., Li, X., & Zhang, B. (2024b). F²Depth: Self-supervised indoor monocular depth estimation via optical flow consistency and feature map synthesis. *Engineering Applications of Artificial Intelligence*, 124(133), Article 108391.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *In CVPR*.
- Hou, Y., Kannala, J., & Solin, A. (2019). Multi-view stereo by temporal nonparametric fusion. *In ICCV*.
- Hu, W., Gao, X., Li, X., et al. (2025). Depthcrafter: Generating consistent long depth sequences for open-world videos[C]. In *Proceedings of the Computer Vision and Pattern Recognition Conference* (pp. 2005–2015).
- Huang, B., et al. (2021). M3VSNet: Unsupervised multi-metric multi-view stereo network. *2021 IEEE international conference on image processing (ICIP)*. IEEE.
- Hui, T.-W. (Jun. 2022). RM-depth: Unsupervised learning of recurrent monocular depth in dynamic scenes. *In Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)* (pp. 1665–1674).
- Im, S., Jeon, H.-G., Lin, S., & So Kweon, I. n (2019). DPSNet: End-to-end deep plane sweep stereo. *ICLR*.
- Johnston, A., & Carneiro, G. (2020). Self-supervised monocular trained depth estimation using self-attention and discrete disparity volume. *In CVPR*.
- Kar, A., Hane, C., & Malik, J. (2017). Learning a multi-view stereo machine. *In NeurIPS*.
- Kendall, A., Martirosyan, H., Dasgupta, S., Henry, P., Kennedy, R., Bachrach, A., & Bry, A. (2017). End-to-end learning of geometry and context for deep stereo regression. *In ICCV*.
- Kingma, D. P., & Adam, J. Ba. (2014). *A method for stochastic optimization*. arXiv: 1412.6980.
- Klingner, M., Termohlen, J.-A., Mikolajczyk, J., & Fingscheidt, T. (2020). Self-supervised monocular depth estimation: Solving the dynamic object problem by semantic guidance. *In ECCV*.
- Klodt, M., & Vedaldi, A. (2018). Supervising the new with the old: Learning SFM from SFM. *In ECCV*.
- Kohli, P., Silberman, N., Hoiem, D., & Fergus, R. (2012). Indoor segmentation and support inference from rgbd images. *In Proc. Eur. Conf. Comput. Vis. (ECCV)*.
- Lee, S., Rameau, F., Pan, F., & Kweon, I. S. (2021). Attentive and contrastive learning for joint depth and motion field estimation. *In Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)* (pp. 4842–4851).
- Li, H., Gordon, A., Zhao, H., Casser, V., & Angelova, A. (2020). Unsupervised monocular depth learning in dynamic scenes. *In CoRL*.
- Li, R., Ji, P., Xu, Y., & Bhanu, B. (2023). MonoIndoor++: Towards better practice of self-supervised monocular depth estimation for indoor environments. *IEEE Transactions on Circuits and System for Video Technology*, 33(2), 830–846.
- Li, R., et al. (2023). Learning to fuse monocular and multi-view cues for multi-frame depth estimation in dynamic scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Liang, Z., Feng, Y., Guo, Y., Liu, H., Chen, W., Qiao, L., Zhou, L.I., & Zhang, J. (2018). Learning for disparity estimation through feature constancy. *In CVPR*.
- Liu, J., Kong, L., Li, B., Wang, Z., Gu, H., & Chen, J. (2025). Mono-ViFi: A unified learning framework for self-supervised single and multi-frame monocular depth estimation. *European conference on computer vision* (pp. 90–107). Cham: Springer.
- Liu, Z., Li, R., Shao, S., Wu, X., & Chen, W. (2023). Self-supervised monocular depth estimation with self-reference distillation and disparity offset refinement. *IEEE Transaction on Circuits and System for Video Technology*, 33(12), 7565–7577.
- Long, X., Liu, L., Theobalt, C., & Wang, W. (2020). Occlusion-aware depth estimation with adaptive normal constraints. *In ECCV*.
- Long, Y., Yu, H., & Liu, B. (Oct. 2022). Two-stream based multi-stage hybrid decoder for self-supervised multi-frame monocular depth. *IEEE Robotics and Automation Letters*, 7 (4), 12291–12298.
- Luo, C., et al. (2020). Every pixel counts ++: Joint learning of geometry and motion with 3D holistic understanding. *IEEE Transactions Pattern Analysis Mach. Intelligent*, 42 (10), 2624–2641.
- Luo, X., Huang, J.-B., Szeliski, R., Matzen, K., & Kopf, J. (2020). Consistent video depth estimation. *In ACM SIGGRAPH*.
- Mahjourian, R., Wicke, M., & Angelova, A. (2018). Unsupervised learning of depth and ego-motion from monocular video using 3D geometric constraints. *In Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* (pp. 5667–5675).
- Martins, D., Van Heck, K., & De Croon, G. (2018). Fusion of stereo and still monocular depth estimates in a self-supervised learning context. *In ICRA*.
- Miao, X., Bai, Y., Duan, H., Huang, Y., Wan, F., Xu, X., Long, Y., & Zheng, Y. (2023). Ds-depth: Dynamic and static depth estimation via a fusion cost volume. *IEEE Transactions on Circuits and Systems for Video Technology*.
- Newcombe, R. A., Izadi, S., & Hilliges, O. (2011). Kinectfusion: Real-time dense surface mapping and tracking. *In UIST*. W.-K.
- V. Patil, C. Sakaridis, A. Liniger, and L. Van Gool, "P3Depth: Monocular depth estimation with a piecewise planarity prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, New Orleans, LA, USA, 2022, pp. 1610–1621.
- Patil, V., Gansbeke, W. V., Dai, D., & Gool, L. V (2020). Don't forget the past: Recurrent depth estimation from monocular video. *In IEEE Robotics and Automation Letters*.
- Ruhkamp, P., Gao, D., Chen, H., Navab, N., & Busam, B. (Dec. 2021). Attention meets geometry: Geometry guided spatial-temporal attention for consistent self-supervised monocular depth estimation. *In Proc. Int. Conf. 3D Vis. (3DV)* (pp. 837–847).
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., et al. (2015). Imagenet large scale visual recognition challenge. *IJCV*.
- Saunders, K., Vogiatzis, G., & Manso, L. J. (2023). Self-supervised monocular depth estimation: Let's talk about the weather. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 8907–8917).
- Shen, D., Ma, F., He, J., Dong, J., Wu, S., & Zhang, Y. (2025). Pose-depth joint optimization and adaptive sampling for self-supervised multi-frame depth estimation. *2025 4th International Symp. Computer Applied Information and Technology (ISCAIT)*.
- Shu, C., Yu, K., Duan, Z., & Yang, K. (2020). Feature-metric loss for self-supervised learning of depth and ego-motion. *In ECCV*.
- Smolyanskiy, N., Kamenev, A., & Birchfield, S. (2018). On the importance of stereo for accurate depth estimation: An efficient semi-supervised deep neural network approach. *In CVPR Workshops*.
- Song, X., Hu, H., Liang, L., Shi, W., Xie, G., Lu, X., & Hei, X. (2023). Unsupervised monocular estimation of depth and visual odometry using attention and depth-pose consistency loss. *IEEE Transaction Multimedia*, 26, 3517–3529.
- Uhrig, J., Schneider, N., Schneider, L., Franke, U., Brox, T., & Geiger, A. (2017). Sparsity invariant CNNs. *In Proc. Int. Conf. 3D Vis. (3DV)* (pp. 11–20).
- Ummenhofer, B., Zhou, H., Uhrig, J., Mayer, N., Ilg, E., Dosovitskiy, A., & Brox, T. (2017). DeMoN: Depth and motion network for learning monocular stereo. *In CVPR*.
- Vaswani, A., Ramachandran, P., Srivivas, A., Parmar, N., Hechtman, B., & Shlens, J. (2021). Scaling local self-attention for parameter efficient visual backbones. *In Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)* (pp. 12889–12899).
- Wang, J., Lin, C., Nie, L., Huang, S., Zhao, Y., Pan, X., & Ai, R. (2024a). Weatherdepth: Curriculum contrastive learning for self-supervised depth estimation under adverse weather conditions. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 4976–4982). IEEE.
- J. Wang, G. Zhang, Z. Wu, X. Li, and L. Liu, "Self-supervised joint learning framework of depth estimation via implicit cues," 2020, arXiv:2006.09876.

- Wang, R., Pizer, S. M., & Frahm, J.-M. (2019a). Recurrent neural network for (un-) supervised learning of monocular video visual odometry and depth. In *CVPR*.
- Wang, X., Zhu, Z., Huang, G., Chi, X., Ye, Y., Chen, Z., & Wang, X. (2023). Crafting monocular cues and velocity guidance for self-supervised multi-frame depth learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(3), 2689–2697.
- Wang, Y., Liang, Y., Xu, H., Jiao, S., & Yu, H. (2024b). Sqlddepth: Generalizable self-supervised fine-structured monocular depth estimation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(6), 5713–5721.
- Wang, Y., Wang, P., Yang, Z., Luo, C., Yang, Y.i., & Xu, W. (2019b). UnOS: Unified unsupervised optical-flow and stereo-depth estimation by watching videos. In *CVPR*.
- Watson, J., et al. (2021). The temporal opportunist: Self-supervised multi-frame monocular depth. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Watson, J., Firman, M., Brostow, G., & Turmukhambetov, D. (2019). Self-supervised monocular depth hints. In *ICCV*.
- Woo, S., Lee, W., Kim, W. J., et al. (2024). ProDepth: Boosting self-supervised multi-frame monocular depth with probabilistic fusion[C]. In *European Conference on Computer Vision* (pp. 201–217).
- Wu, G., Liu, H., Wang, L., Li, K., Guo, Y., & Chen, Z. (2024). Self-supervised multi-frame monocular depth estimation for dynamic scenes. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(6), 4989–5001.
- Wu, J., Li, R., Xu, H., Zhao, W., Zhu, Y., Sun, J., & Zhang, Y. (2024). GoMVS: Geometrically consistent cost aggregation for multi-view stereo. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 20207–20216).
- Yang, G., et al. (2025a). Multi-view stereo with geometric encoding for large-scale dense scene reconstruction. *IEEE Transactions on Automation Science and Engineering*, 22, 22568–22586.
- Yang, G., et al. (2025b). Toward end-to-end underwater multi-view stereo for real-world dense scene reconstruction. *IEEE Transactions on Geoscience and Remote Sensing*, 63, 1–19. Art no. 4214019.
- Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., & Zhao, H. (Jun. 2024). Depth anything: Unleashing the power of large-scale unlabeled data. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)* (pp. 10371–10381).
- Yao, Y., Luo, Z., Li, S., Fang, T., & Quan, L. (2018). Mvsnet: Depth inference for unstructured multi-view stereo. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 767–783). pages.
- Yin, W., Chen, H., et al. (2023). Metric3d: Towards zero-shot metric 3d prediction from a single image[C]. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9043–9053).
- Yin, W., Zhang, J., Wang, O., Niklaus, S., Chen, S., Liu, Y., & Shen, C. (2022). Towards accurate reconstruction of 3d scene shape from a single monocular image. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Yin, Z., & Shi, J. (2018). GeoNet: Unsupervised learning of dense depth, optical flow and camera pose. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 1983–1992.
- Y. Yu, Z. Zheng, D. Lian, Z. Zhou, and S. Gao, “Single-image piecewise planar 3D reconstruction via associative embedding,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Long Beach, CA, USA, 2019, pp. 1029–1037.
- Zhan, H., Garg, R., Saroj Weerasekera, C., Li, K., Agarwal, H., & Reid, I. (2018). Unsupervised learning of monocular depth estimation and visual odometry with deep feature reconstruction. In *CVPR*.
- Zhang, F., Qi, X., Yang, R., Prisacariu, V., Wah, B., & Torr, P. (2020). Domain-invariant stereo matching networks. In *ECCV*.
- Zhang, H., Shen, C., Li, Y., Cao, Y., Liu, Y.u., & Yan, Y. (2019). Exploiting temporal consistency for real-time video depth estimation. In *ICCV*.
- Zhang, Y., Gong, M., Zhang, M., & Li, J. (Dec. 2024). Self-supervised monocular depth estimation with Self-perceptual anomaly handling. *IEEE Transactions on Neural Networks and Learning Systems*, 35(12), 17292–17306.
- Zhang, Y., Gong, Y., Li, M., Zhang, J., Fang, J., & Zhao, H. (2022). Self-supervised monocular depth estimation with multiscale perception. *IEEE Transactions On Image Processing : A Publication of the IEEE Signal Processing Society*, 31, 3251–3266.
- Zhao, C., et al. (2022). MonoViT: Self-supervised monocular depth estimation with a vision transformer. In *Proc. Int. Conf. 3D Vis. (3DV)* (pp. 668–678).
- H. Zhao, S. Liu, S. Fu, Y. Liu, and T. Yu, “Towards better generalization: Joint depth-pose learning without Posenet,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Seattle, WA, USA, 2020, pp. 9151–9161.
- Zhou, T., Brown, M., Snavely, N., & Lowe, D. (2017). Unsupervised learning of depth and ego-motion from video. In *CVPR*.